OVERVIEW

Team

The most examined chapter in SE423: both 251 midterms took most of their short questions from it, and both finals have a whole team-based collaboration question. It has three parts. Team development: Tuckman's stages, team objectives and structures, and the team needs pyramid (trust → managed conflict → commitment → accountability → results) with RACI and skills matrices as the accountability tools. Team models: business, democratic, chief-programmer, skunk works, SWAT, and why large teams need hierarchy (n(n−1)/2 channels). Managing people: motivation and personality types, cohesiveness and groupthink, staff selection, working environments, shaping project culture, technical vs. adaptive leadership, and reaching consensus.

The instructor's style: questions ask you to apply an idea to a team situation ("team members disagree: how do you encourage collaboration?"). Saying one word ("negotiation") is not enough; explain what you do, how, and why.

Team pyramid

Trust → managed conflict → commitment → accountability → results. Each layer needs the one below; trust is the foundation.

Tuckman

Forming → storming (competing for leadership) → norming (agree how to work) → performing → adjourning.

RACI

R does the work, A owns the outcome (exactly one per task), C gives input (two-way), I is kept updated (one-way).

Team models

Business (tech lead + equals, most common), democratic (whole team decides), chief-programmer (surgical), skunk works, SWAT.

Leadership

Technical leadership for problems solvable by known means; adaptive leadership for new or complex ones. Most IT projects need both.

Consensus

Not unanimous agreement: everyone feels heard and the group moves forward committed. Persuade, don't dictate.

HOW THE LECTURE IS ORGANIZED

Slides 1–9 Team development: influence, the pyramid, Tuckman, team objectives

Slides 10–24 The pyramid layer by layer: trust, conflict, commitment, accountability (RACI, RAM, skills matrix), results

Slides 25–34 Team models and team size

Slides 35–56 Managing people: motivation, personality types, cohesiveness, communication, selection, environment

Slides 57–70 Project culture, managing thought workers, making people better

Slides 71–77 Leadership, consensus and key points

EXAM FOCUS

Midterm 251 (both versions): MCQs on Tuckman's storming stage, the foundation of teamwork and the advantage of a cohesive team; short questions on good-team characteristics, vulnerability-based trust, healthy conflict, the best time to reinforce accountability, business vs. democratic teams, technical vs. adaptive leadership and a RACI table; a case on accountability after an untested release. Finals: commitment, trust and conflict, accountability and transparency (Final 242 Q5), the communication-channel formula (Final 242), a CI/CD consensus decision and a milestone celebration (Final 251 Q5). The instructor's 2026 midterm asked how a PM encourages collaboration when members disagree.

HOW TO READ THIS PAGE

Every original slide is shown unchanged. Yellow marks on a slide point to a numbered note: hover or tap a mark to jump to its note, and hover a note to see its marks. Questions tagged **Past exam** appeared on a real paper; **Prediction** questions are likely angles, not confirmed. Pick an answer to each question: a wrong pick tells you why it is wrong, so you can try again.

SLIDE 1 / 77 **Slide 1**



Team

SE423: Software Project Management

1

NOTES

1 The deck has three parts: **team development** and the team needs pyramid (slides 3–24), **team models** (25–34) and **managing people** (35–77). The pyramid (trust → conflict → commitment → accountability → results) is the spine of the chapter and of most exam questions on it.

SLIDE 2 / 77 **Outline**

Outline

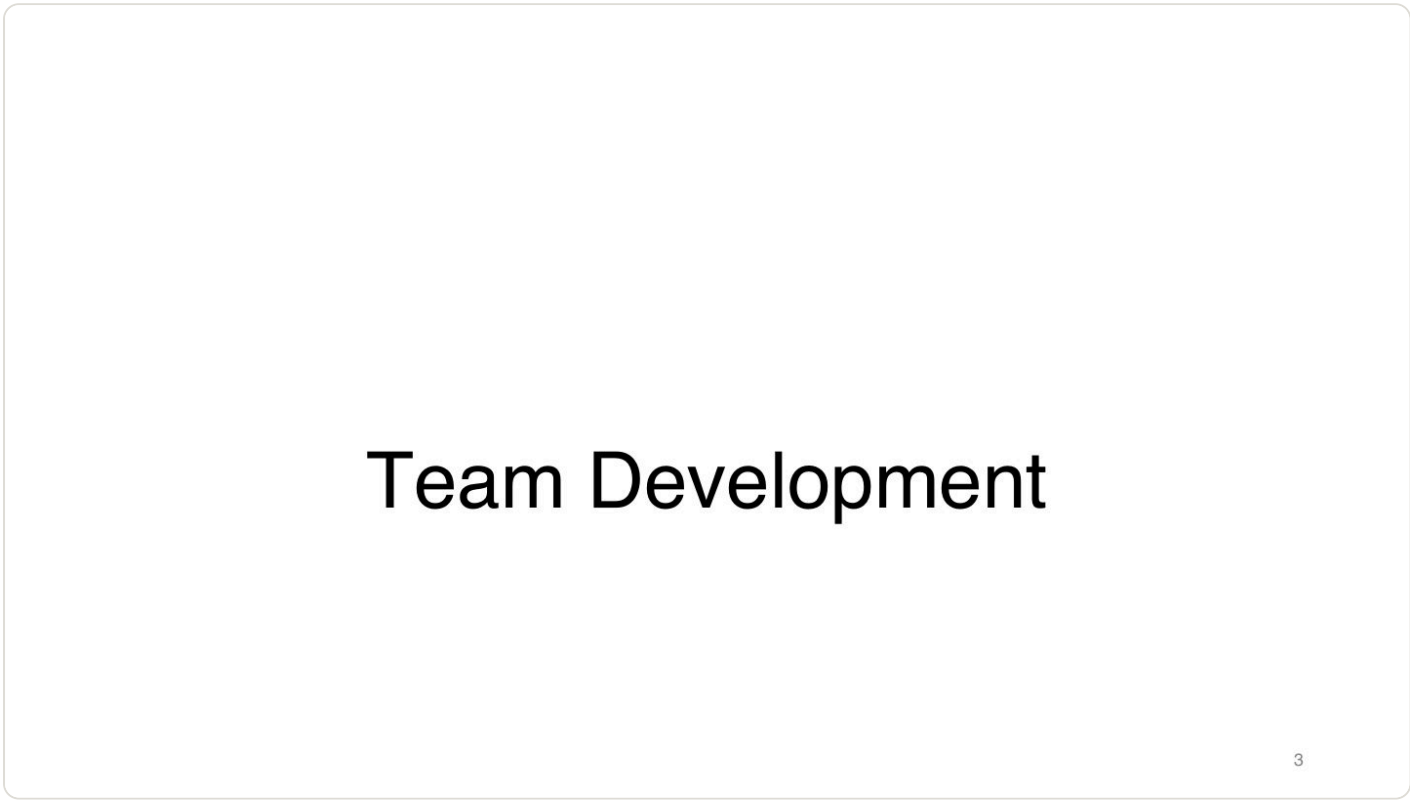
- Team Development
- Building Trust
- Managing Conflict
- Embracing Accountability
- Team Models
- Managing People

2

NOTES

1. **Team development** first: how a group becomes a team.
2. Then three of the pyramid layers in order: **building trust**, **managing conflict**, **embracing accountability** (commitment and results are covered between them).
3. Then the **team models** (how teams are organized) and **managing people** (motivation, selection, culture, leadership).

SLIDE 3 / 77 **Slide 3**



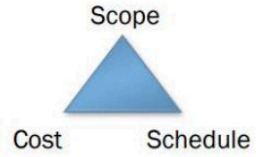
NOTES

1. Section divider: **Team Development**.

 Team Development - Introduction

Team Development - Introduction

- What are your responsibilities as a team member?
- As a Project Manager?
- How does Team Development relate to the Project Triangle?



- What's more important? The strength of individual team members, or the strength of the team?
- What are some characteristics of a good team?

4

NOTES

1. Discussion questions to open the topic. As a **team member** you own your tasks and support others; as a **PM** you build the conditions for the team to work: clear goals, roles, trust.
2. The link to the **project triangle**: a weak team costs you on scope, schedule or cost. People are the resource that turns the plan into results.
3. Individuals vs. team: a team of strong individuals who don't trust each other usually loses to a well-functioning team of average people. Team strength wins.
4. "Characteristics of a good team" was a real midterm short question. Answer it with the pyramid, from the bottom up.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q1 — Q1

What are key characteristics of a good team?
Answer: Members trust each other (the foundation), so they speak openly and admit mistakes; they have healthy conflict about ideas rather than people; they commit to decisions with clarity and buy-in; they hold each other accountable; and they put team results ahead of individual credit. A good team also shares a clear goal with clear roles, communicates openly, is cohesive, balances personality types and is the right size (about 4–6 developers plus a lead).

 Quote

Quote

"Software projects fail for one of two general reasons: the project team lacks the knowledge to conduct a software project successfully, or the project team lacks the resolve (*determination, commitment, mental toughness*) to conduct a project effectively."

- Steve C McConnell, Author of "code complete"

NOTES

1. McConnell's two reasons projects fail. **Knowledge**: the team doesn't know how to run a software project. Training, hiring and process fix this.
2. **Resolve** (determination, commitment, mental toughness): the team knows how but doesn't push through. This is what team building, motivation and leadership fix, and it is what the rest of the chapter is about.

 Team Skills, Influence, and Style

Team Skills, Influence, and Style

- Skills

Skill	Example Activity
Planning	Which resources are pre-assigned to the project?
Negotiation	Getting the best possible resources. Sharing resources.
Hiring	Outsourcing, virtual teams
Risk Management	What if resources become unavailable?
Judgment	Is it possible assumptions are being made regarding skill levels?

- Types of Influence

Formal (Legitimate)	Reward	Penalty (Coercive)	Expert	Referent
Authority granted by organizational hierarchy or role.	Influence through the ability to offer incentives	Influence through the threat or use of punishment.	Influence based on specialized knowledge or skills.	Influence based on personal traits, charisma, or likability.

- Leadership Styles

Directing	Facilitating	Coaching	Supporting
Autocratic	Consultative	Consultative- Autocratic	Consensus
Delegating	Bureaucratic	Charismatic	Democratic / Participative
Laissez-Faire	Analytical	Driver	Influencing

6

NOTES

1. The **types of influence** a PM can use. **Formal (legitimate)**: authority granted by the organizational hierarchy or role. The skills table above lists the PM's team skills (planning, negotiation, hiring, risk management, judgement) and the bottom table lists leadership styles (directing, facilitating, coaching, supporting, etc.).
2. **Reward**: influence through the ability to offer incentives (bonuses, interesting assignments).
3. **Penalty (coercive)**: influence through the threat or use of punishment. It gets compliance, not commitment.
4. **Expert**: influence based on specialized knowledge or skills.
5. **Referent**: influence based on personal traits, charisma or likability. Formal, reward and penalty come from your **position**; expert and referent come from **you**, and they last longer because people choose to follow.

EXAM ANGLE

Prediction Q1

A senior developer persuades the team because everyone respects her deep database knowledge. This is:

A Reward influence

B Formal influence

C **Expert influence**

D Penalty influence

Prediction Q2

Which types of influence come from the person rather than from the position?

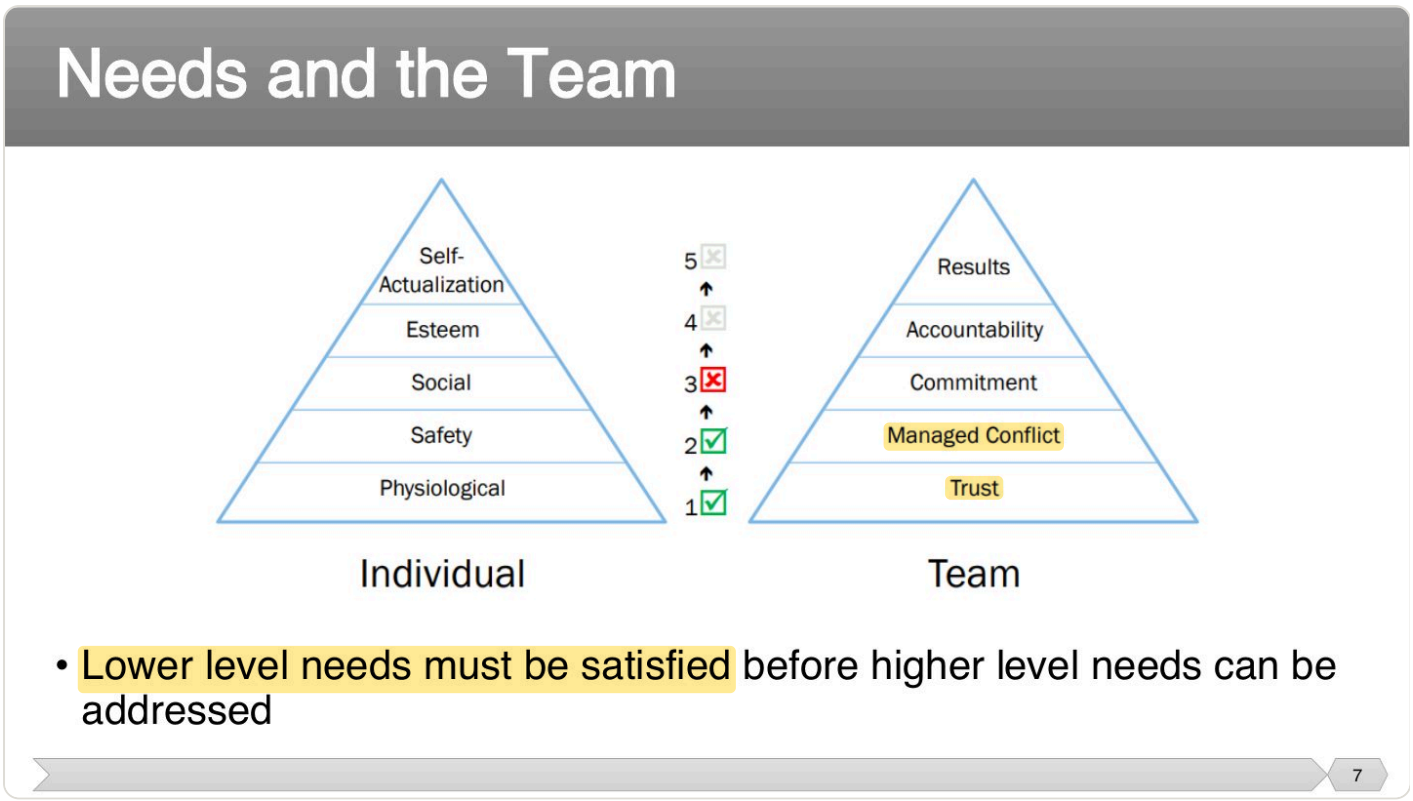
A **Expert and referent**

B Reward and penalty

C Formal and penalty

D Formal and reward

 Needs and the Team



NOTES

1. The rule from Maslow: **lower-level needs must be satisfied before higher ones** can be addressed. On the left is Maslow's individual pyramid; on the right the **team needs pyramid** (Lencioni).

2. The team pyramid from the bottom: **trust**, then **managed conflict**, then **commitment**, then **accountability**, then **results**. You can't have healthy conflict without trust, or commitment without having debated the decision.

3. **Results** sit at the top: the team's collective goals. The ticks on the slide show a team that has secured trust and conflict but is stuck at commitment, so the PM works on that layer next.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 8 (also V2 MCQ 8) Q1

Which of the following is considered the foundation of teamwork?

A Accountability

B **Trust**

> Trust is the base of the team pyramid: trust, managed conflict, commitment, accountability, results.

C Commitment

D Consensus

Prediction Q2

In the team needs pyramid, which layer sits directly above trust?

A Commitment

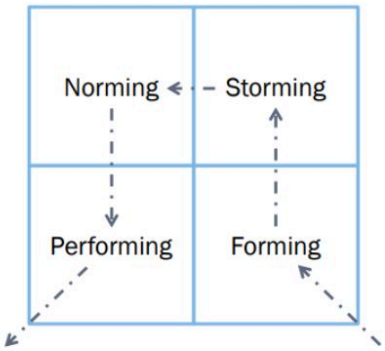
B Accountability

C **Managed conflict**

D Results

 Tuckman's Model for Team Development

Tuckman's Model for Team Development



1. Forming: The group comes together and gets to initially know one another and form as a group.
2. Storming: A chaotic competing for leadership and trials of group processes (compete for influence or control, often without clear structure or authority)
3. Norming: Eventually agreement is reached on how the group operates
4. Performing: The group practices its craft and becomes effective in meeting its objectives
5. Dissolving/Adjourning

NOTES

1. **Tuckman's model**. **Forming**: the group comes together and members get to know each other; they are polite and depend on the leader.
2. **Storming**: a chaotic competing for leadership and trials of group processes; members compete for influence or control, often without clear structure or authority. The PM's job here is to surface and manage the conflict, not suppress it.
3. **Norming**: agreement is reached on how the group operates (roles, norms, ways of working).
4. **Performing**: the group practices its craft and becomes effective at meeting its objectives.
5. **Adjourning**: the team dissolves at the end of the project. Order: form → storm → norm → perform → adjourn.

TIPS & TRICKS

- The order matters, and they rhyme: **F**orm, **S**torm, **N**orm, **P**erform, then **A**djourn.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 7 (also V2 MCQ 7) Q1

According to Tuckman's model of team development, which stage is characterized by members competing for leadership and struggling with group processes?

A Storming

> Storming is when members compete for leadership and test group processes; norming comes after, when agreement is reached.

B Norming

C Adjourning

D Performing

Prediction Q2

The stage where the team reaches agreement on how it will operate is:

A Performing

B Norming

C Forming

D Storming

 Team Structure

Team Structure

* She took her time explaining this.

- What's the team's objective? **three distinct types of team objectives** in project management, each shaping the team's structure, behavior, and required

objective	Structure Implication
Problem resolution: Complex, poorly-defined problem. Focuses on 1-3 specific issues. Ex: fixing a showstopper defect (that stop progress entirely) • Team traits: Sense of urgency and pressure	Flexible roles, rapid decision-making, and possibly a temporary task force
Creativity: New product development: Generating new ideas, products, or approaches. • Team Traits: Encourages divergent thinking and experimentation, tolerates ambiguity and iteration.	Looser hierarchy, diverse skill sets, and a culture that supports risk-taking and idea sharing
Tactical execution: Carrying-out well-defined plan. Focused tasks and clear roles. • Team Traits: Clear roles and responsibilities. Strong emphasis on timelines, coordination, and accountability.	Hierarchical or matrix structure with tight controls and performance tracking.

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NOTES

1. The team's **objective** shapes its structure. **Problem resolution**: a complex, poorly defined problem focused on 1–3 issues (e.g. a showstopper defect). Traits: urgency and pressure. Structure: flexible roles, rapid decision-making, possibly a **temporary task force**.

2. **Creativity**: new product development, generating new ideas. Traits: divergent thinking, experimentation, tolerance of ambiguity. Structure: **looser hierarchy**, diverse skills, a culture that supports risk-taking.

3. **Tactical execution**: carrying out a well-defined plan. Traits: clear roles, timelines, coordination, accountability. Structure: hierarchical or matrix with **tight controls** and performance tracking. (The instructor noted she spent time explaining this slide.)

EXAM ANGLE

Past exam · Midterm 251 · V1 · Case Q4 (also V2) Q1

As the project manager, you must establish the team structure for this project. What team structure would you create to ensure smooth development, stakeholder engagement, and integration with Edugate, and why?

Answer: A cross-functional Agile team: a Project Manager, a Product Owner from Edugate/IT, developers, a UI/UX designer, QA testers, a faculty advisor representative and a student representative (1.0). This puts every stakeholder's voice inside the team, balances usability against compliance, and supports iterative delivery (0.5).

Prediction Q2

A team must roll out an already-designed module on a fixed timeline. Which objective type and structure fit?

Answer: Tactical execution: carrying out a well-defined plan. Use a hierarchical or matrix structure with clear roles, tight controls and performance tracking, because the work is known and the risk is in coordination and deadlines, not in discovering what to build.

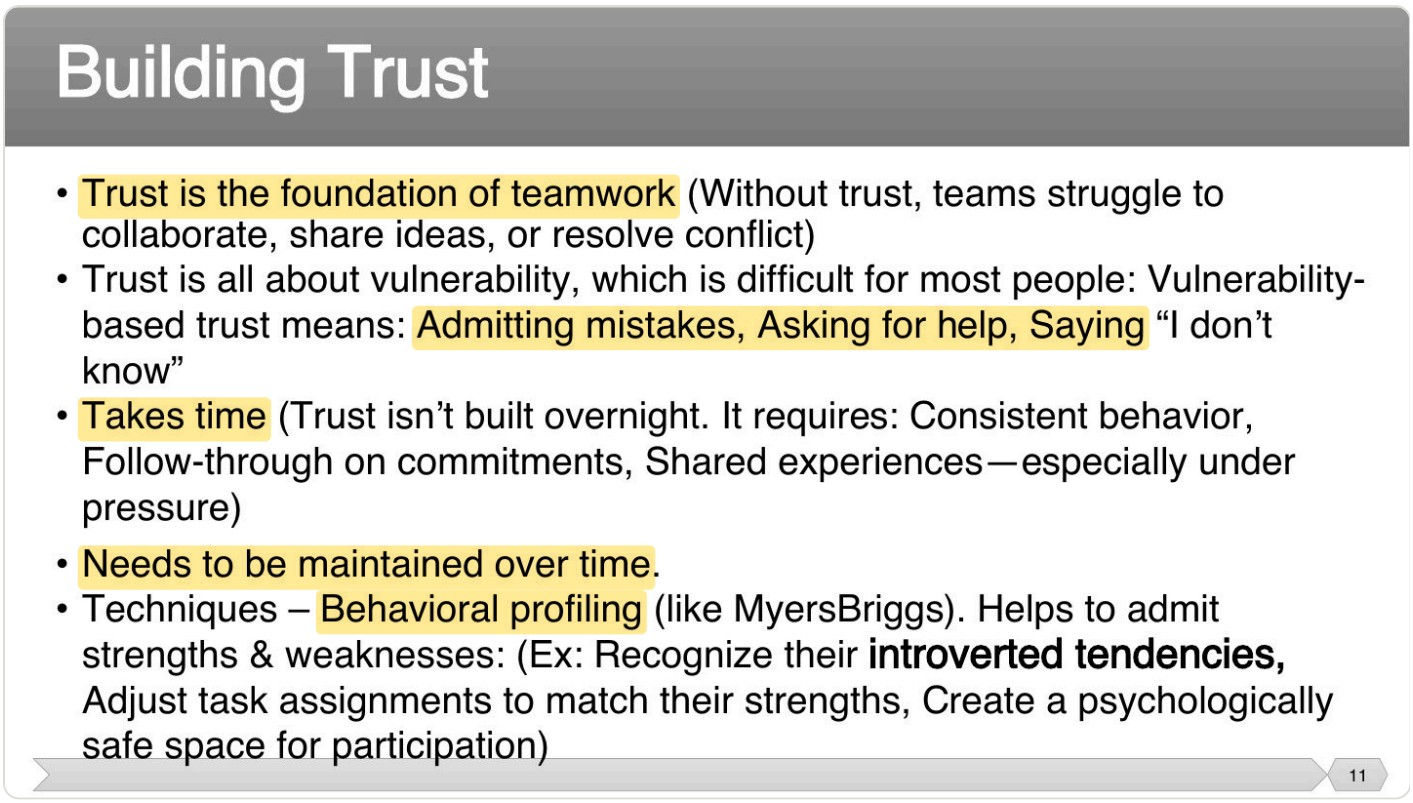
SLIDE 10 / 77 **Slide 10**



NOTES

1 Section divider: **Building Trust**, the base of the pyramid.

 Building Trust



NOTES

1. **Trust is the foundation of teamwork**: without it teams struggle to collaborate, share ideas or resolve conflict.

2. Trust is about **vulnerability**. **Vulnerability-based trust** means admitting mistakes, asking for help and saying "I don't know" without fear it will be used against you. This list was a real midterm question.

3. It **takes time** (consistent behavior, follow-through on commitments, shared experiences under pressure) and must be **maintained**: one broken promise can undo months.

4. A technique: **behavioral profiling** (e.g. Myers-Briggs) helps people admit strengths and weaknesses openly; the PM can adjust assignments to match strengths and create a psychologically safe space.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q2 (also V2 Short Q1) Q1

What does vulnerability-based trust involve?

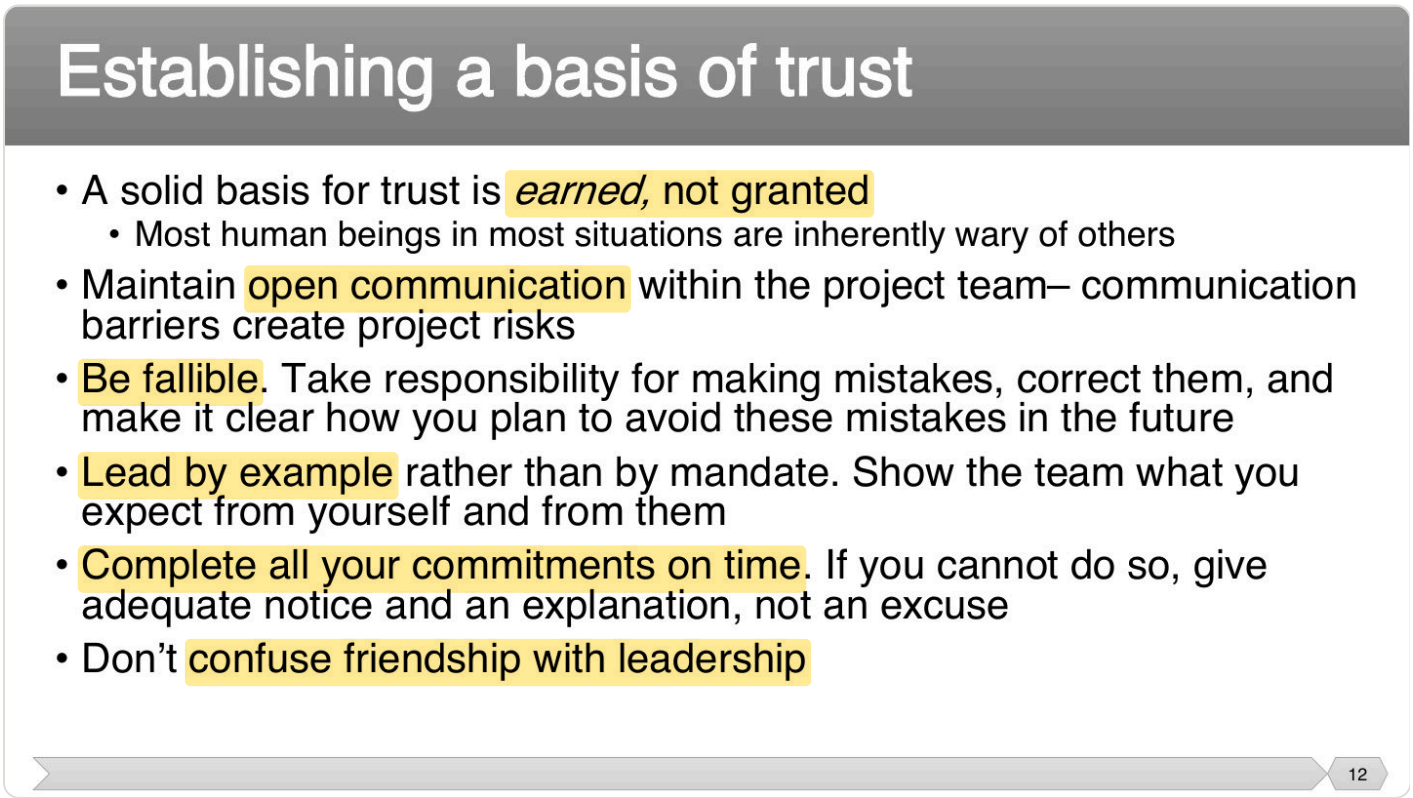
Answer: Team members are comfortable being open with each other: admitting mistakes and weaknesses, asking for help, and saying "I don't know", without fear that it will be used against them. It takes time to build through consistent behavior, follow-through and shared experiences under pressure, must be maintained, and is the foundation of the team pyramid.

Past exam · Final 242 · Q5.2 Q2

How does fostering trust and managing conflict within a software project team contribute to improved collaboration and project success?

Answer: Trust lets members admit mistakes, share information and ask for help, so problems surface early and people cooperate instead of protecting themselves. Managed (healthy) conflict means ideas are debated openly and respectfully, which gives better technical decisions and avoids groupthink or disputes that turn personal. Together they lead to real commitment and collaboration, which drives quality and project success.

SLIDE 12 / 77 **Establishing a basis of trust**



NOTES

1. Trust is **earned, not granted**: a title doesn't make people trust you, and most people are naturally wary of others.
2. Keep **open communication**: communication barriers are project risks because problems stay hidden.

3. **Be fallible**: own your mistakes, correct them and explain how you'll avoid them in future. This models the vulnerability you want from the team.
4. **Lead by example**, not by mandate: show the standard you expect.
5. Keep **commitments**; if you can't, give notice and an explanation, not an excuse.
6. **Friendship ≠ leadership**: warmth helps, but leading needs clarity and boundaries (repeated on slide 76).

EXAM ANGLE

Prediction Q1

Which behavior does the slide recommend for establishing trust?

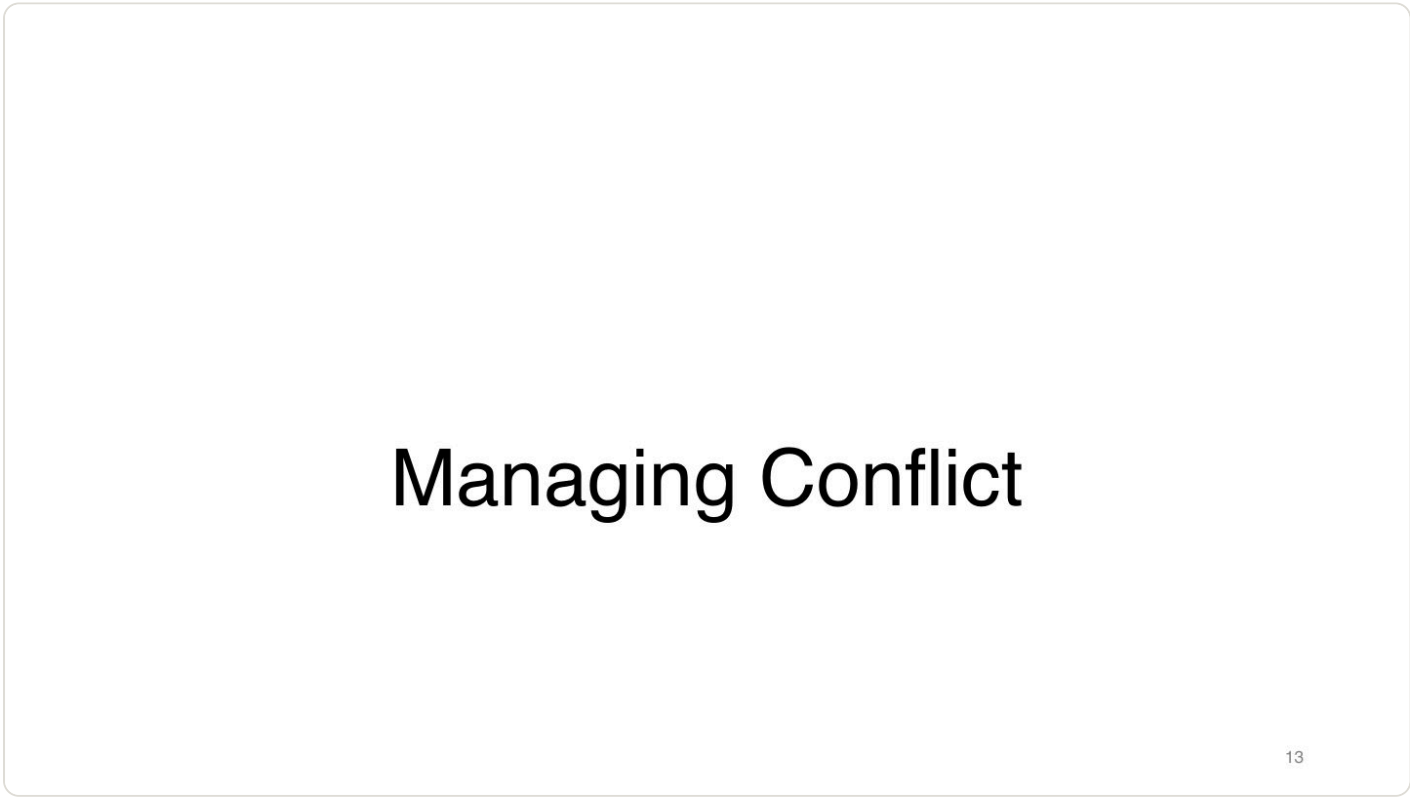
A **Admit mistakes and explain how you will avoid them**

B Give excuses when commitments slip

C Lead by mandate

D Hide mistakes to protect your authority

SLIDE 13 / 77 **Slide 13**



NOTES

1. Section divider: **Managing Conflict**, the second layer.

SLIDE 14 / 77 **Managing Conflict**

Managing Conflict

- Good (healthy) conflict among team members requires trust, which is all about engaging in unfiltered, passionate debate around issues.
- Even among the best teams, conflict will at times be uncomfortable.
- Conflict norms will vary in each team, and must be discussed and made clear:
  - Norms = rules of engagement: These define what's acceptable in debate—tone, language, timing, emotional expression.
- The fear of occasional personal conflict should not deter a team from having regular, productive debate:
  - Avoidance is costly: Teams that fear conflict often fall into **groupthink** (psychological phenomenon where a cohesive group prioritizes harmony and consensus over critical thinking and realistic decision-making), **suppress dissent** (actively prevent or silence disagreement, criticism, or opposition), or make poor decisions.

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NOTES

1. **Healthy conflict** is unfiltered, passionate debate around **issues**, not people, and it requires trust underneath.
2. Even in the best teams it will sometimes be **uncomfortable**; that is normal, not a sign of failure.
3. **Conflict norms** are the team's **rules of engagement** (tone, language, timing, emotional expression); they differ per team and must be discussed and made explicit.
4. **Avoidance is costly**: teams that fear conflict fall into **groupthink** (harmony over critical thinking), suppress dissent and make poor decisions nobody challenged.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q3 (also V2 Short Q3) Q1

Why is healthy conflict important in teams?

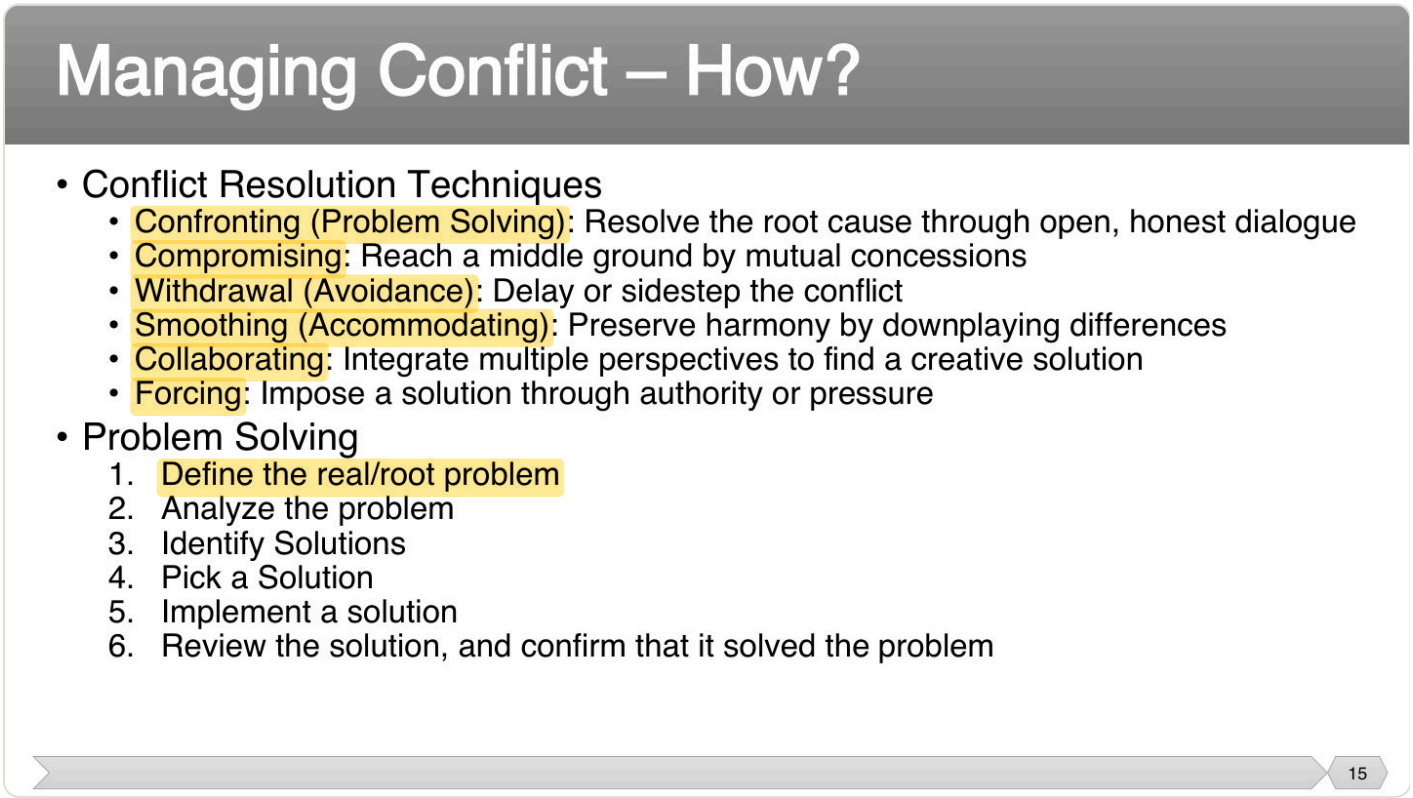
Answer: Healthy conflict is open debate about ideas, not people. It brings out different perspectives so the team makes better decisions, avoids groupthink and artificial harmony, and surfaces problems early. Because everyone has been heard, members are more willing to commit to the final decision even if they disagreed. It needs trust underneath and agreed conflict norms.

Prediction Q2

A team values harmony so much that it stops critically evaluating ideas. This is:

- **A Groupthink**
- B Accountability
- C Storming
- D Healthy conflict

SLIDE 15 / 77 **Managing Conflict – How?**



NOTES

1 **Confronting (problem solving)**: resolve the root cause through open, honest dialogue. Gives the most lasting result.

2. **Compromising**: both sides give something up to reach a middle ground. Quick, but nobody gets what they needed.
3. **Withdrawal** delays or sidesteps the conflict; **smoothing** keeps harmony by downplaying differences. Both leave the cause in place.
4. **Collaborating**: integrate multiple perspectives into a creative solution. With confronting, the preferred approach for a team.
5. **Forcing**: impose a solution through authority or pressure. Fast in an emergency, but it costs buy-in.
6. The **problem-solving steps**: define the real/root problem → analyze it → identify solutions → pick one → implement it → review that it solved the problem.

TIPS & TRICKS

- Sentence, in slide order: **C**ool **C**ats **W**ill **S**till **C**alm **F**ights. Confronting, Compromising, Withdrawal, Smoothing, Collaborating, Forcing.

EXAM ANGLE

Past exam · Midterm 261 · Short — Q1

Team members disagree about the best solution to a problem. How can the project manager encourage collaboration? (Saying "negotiation" alone is not enough: explain.)
Answer: Treat it as healthy conflict and use collaborating/confronting (problem solving) rather than forcing or smoothing. Bring the people together for an open discussion focused on the problem, not on people; define the real problem and agree the criteria for a good solution; ask each person to present their option with its reasoning and data; weigh the pros and cons of each option together; then converge on the option that best serves the project, or combine ideas into a better one. Work with those who preferred other options so they buy in. This gives a better decision and real commitment because everyone was heard.

Prediction — Q2

A PM ends a debate by declaring "we use my design, end of discussion." Which technique is this?

A Compromising

B Smoothing

C **Forcing**

D Confronting

Managing and resolving conflict

- Every team encounters conflict: successful teams resolve conflicts, while unsuccessful teams force conflict to the background
- Disagreement is not the same as conflict: conflict involves a hardening of position and associated intractability (insolvability)
- Virtually all conflicts can be traced to a clash between two parties
- To preserve a team's effectiveness, those outside the conflict must assume a neutral stance, even if they themselves agree with one of the parties to the conflict
- One of those outside the conflict must take on the role of conciliator; he or she must mediate between the two parties to defuse the situation
- The goal is to reduce the conflict to a disagreement which can then be addressed through consensus
- Strive for reconciliation between the conflicting parties, if possible

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NOTES

1. Every team meets conflict: successful teams **resolve** it, unsuccessful ones push it into the background where it festers.
2. **Disagreement ≠ conflict**. A disagreement is about an idea; a conflict involves a **hardening of position** and intractability (nobody will move). Almost every conflict traces back to a clash between two parties.
3. Those outside the conflict stay **neutral**, even if they agree with one side, and one of them acts as **conciliator**, mediating to defuse the situation. The goal is to reduce the conflict back to a disagreement that can be settled by consensus.
4. Finally, aim for **reconciliation** between the two parties so the working relationship survives.

EXAM ANGLE

Prediction Q1

Disagreement and conflict are the same thing.

T True

F False

> Conflict involves a hardening of position and intractability; the goal is to reduce it back to a disagreement.

Achieving Commitment

- Commitment requires **clarity** & **buy-in**:
  - Clarity ensures everyone knows exactly what was decided, why it matters, and what's expected next.
  - Buy-in means team members support the decision—even if they didn't fully agree—because they trust the process and feel heard.
- Clarity requires that teams avoid assumptions and ambiguity, and end discussions with a clear understanding about what they've decided upon.
- Buy-in does not require consensus. Members of great teams learn to disagree with one another and still commit to a decision.

NOTES

1. **Commitment** (the third layer) requires **clarity and buy-in**.
2. **Clarity**: everyone knows exactly what was decided, why it matters and what is expected next. Teams avoid assumptions and ambiguity and end each discussion with an explicit decision.
3. **Buy-in**: members support the decision even if they didn't fully agree, because they trust the process and felt heard (this is why healthy conflict comes first).
4. **Buy-in does not require consensus**: great teams can disagree and still commit ("disagree and commit"). A common trap: "commitment requires everyone to agree" is false.

TIPS & TRICKS

- **CuB**: Clarity, Buy-in.

EXAM ANGLE

Past exam · Final 242 · Q5.1 — Q1

In the context of the Team Needs Pyramid, explain how a project manager can foster commitment among team members.

Answer: In the pyramid (trust → managed conflict → commitment → accountability → results), commitment sits on trust and healthy conflict: people commit when they trust each other and have had their say. The PM fosters it with clarity (clear decisions, goals, roles and deadlines, confirmed at the end of each meeting, with no assumptions) and buy-in (letting everyone voice their views so they support the decision even if they disagreed; consensus isn't required). Recording decisions and owners (e.g. in a RACI) then lets commitment be followed by accountability.

Prediction — Q2

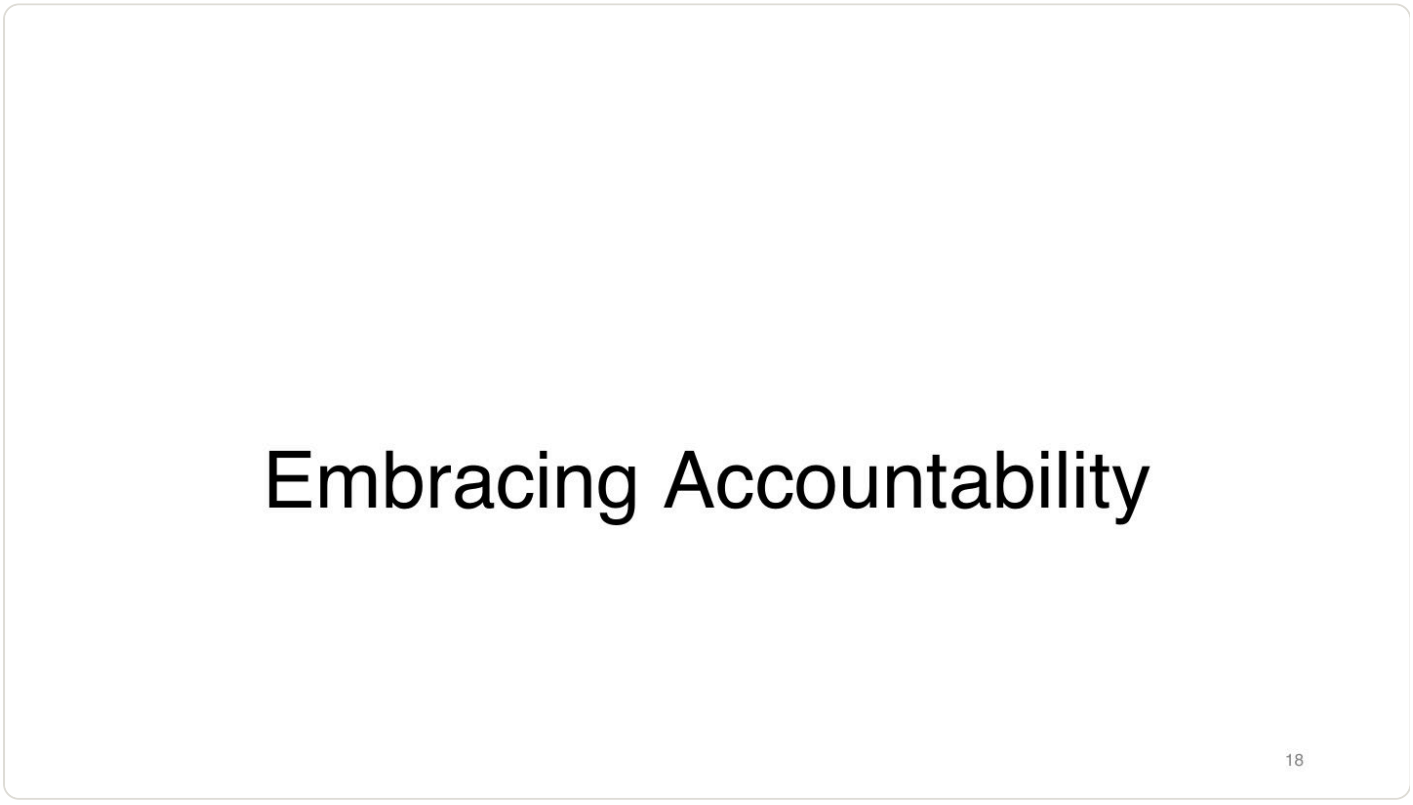
Team buy-in requires that every member agrees with the decision.

T True

F False

> Buy-in does not require consensus.

SLIDE 18 / 77 Slide 18



NOTES

1. Section divider: **Embracing Accountability**, the fourth layer.

 Embracing Accountability

Embracing Accountability

- Accountability on a strong team occurs directly among peers: In high-performing teams, members don't just rely on the manager to enforce standards—they hold each other accountable.
- For a culture of accountability to thrive, a leader must demonstrate a willingness to confront accountability: A project manager should model transparency, follow through on commitments, and address underperformance respectfully but firmly.
- Best opportunity occurs during meetings and regular review of accomplishments. Example: During a sprint review, the PM might say, "*We committed to five stories but completed three. Let's explore what happened and how we can improve next sprint.*"

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NOTES

1. On a strong team accountability happens **directly among peers**: members hold each other to standards instead of waiting for the manager.
2. The leader must show a **willingness to confront accountability**: be transparent, follow through, and address underperformance respectfully but firmly. If the PM avoids it, so will the team.
3. The **best opportunity** is during **meetings and regular reviews** of accomplishments, e.g. at a sprint review: "We committed to five stories but completed three. Let's explore what happened and how we can improve." Note the tone: about the commitment, not blaming a person.

EXAM ANGLE

Past exam · Midterm 251 · V2 · Short Q2 Q1

What's the best time to reinforce accountability?
Answer: During meetings and regular reviews of accomplishments, such as sprint reviews and status meetings, where what was committed is compared with what was done (e.g. "we committed to five stories but completed three; let's explore why"). Make commitments clear and public at the start and address a missed commitment when it happens rather than waiting for the end of the project, so peers hold each other accountable continuously.

Past exam · Midterm 251 · V1 · Case Q1 (also V2) Q2

During a sprint deployment, a key feature was discovered to be not thoroughly tested and caused issues after release. As the project manager, what steps would you follow to address this issue and ensure accountability within the team?
Answer: Use a RACI matrix to establish who was Responsible, Accountable, Consulted and Informed for testing that feature (0.5). Hold a retrospective to find the root cause (0.25). Make the responsible roles clear without blaming individuals (0.25). Take corrective action: release a hotfix, redistribute tasks, and add automated testing so it can't recur (0.25). Inform the stakeholders of the recovery plan (0.25).

Past exam · Final 242 · Q5.3 Q3

Your software project team is experiencing delays due to unclear task ownership and inconsistent updates. As the project manager, how would you apply a team-based approach to improve accountability and transparency within the team? Describe specific steps you would take to enhance collaboration and increase the project's chances of success.
Answer: 1) Clarify ownership with a RACI matrix so every task has a Responsible person and exactly one Accountable person. 2) Hold short daily stand-ups (done, next, blocked) so updates are consistent. 3) Put all tasks on a shared board (e.g. Jira or Kanban) with visible goals and deadlines so progress is transparent. 4) Review commitments against results in regular reviews, encourage peer accountability, and run retrospectives to fix what caused the delays.

 Embracing Accountability

Embracing Accountability

- RACI – Responsible, Accountable, (Support), Consulted, Informed

	Project Manager	Project Sponsor	Developer 1	Tester 1
Activity 1	A	I	R	
Activity 2	R	S		
Activity 3	C		R	S
Activity 4	RA			C
Activity 5	A	R	S	

Note: "Support" is not part of the original RACI acronym but is sometimes ad
in variants like RASCI to reflect collaborative roles more accurately.

NOTES

1. **RACI**: **R**esponsible does the work; **A**ccountable owns the outcome and signs off (exactly **one** A per activity); **C**onsulted gives input before or during the work (two-way); **I**nformed is kept updated (one-way). Several R's on one activity are fine.
2. **Support** (the S in RASCI variants) is not in the original acronym; it marks people who help the Responsible person.
3. In the example, Activity 4 has the project manager as **RA**: one person can be both Responsible and Accountable for a task they do and own.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q6 (V2 asked T2 and T5) Q1

Your team has been assigned to develop a mobile app for a local fitness center. The app will allow users to book classes, track workouts, and receive personalized fitness tips. The project is expected to be completed in 8 weeks. Fill in the RACI assignments for T1 and T6 only.

Answer: These cells come from the instructor's key. The person who does the work is R (the client representative for requirements, the UI/UX designer for the interface, the lead developer for building and launching, the marketing specialist for promotion). Whoever's expertise is needed is C, and everyone else is I. The key makes the PM Accountable (A) for requirements and launch, the two tasks that commit the project; on the other tasks the PM is kept Informed.

Prediction Q2

In RACI, who owns the outcome and signs off on the task?

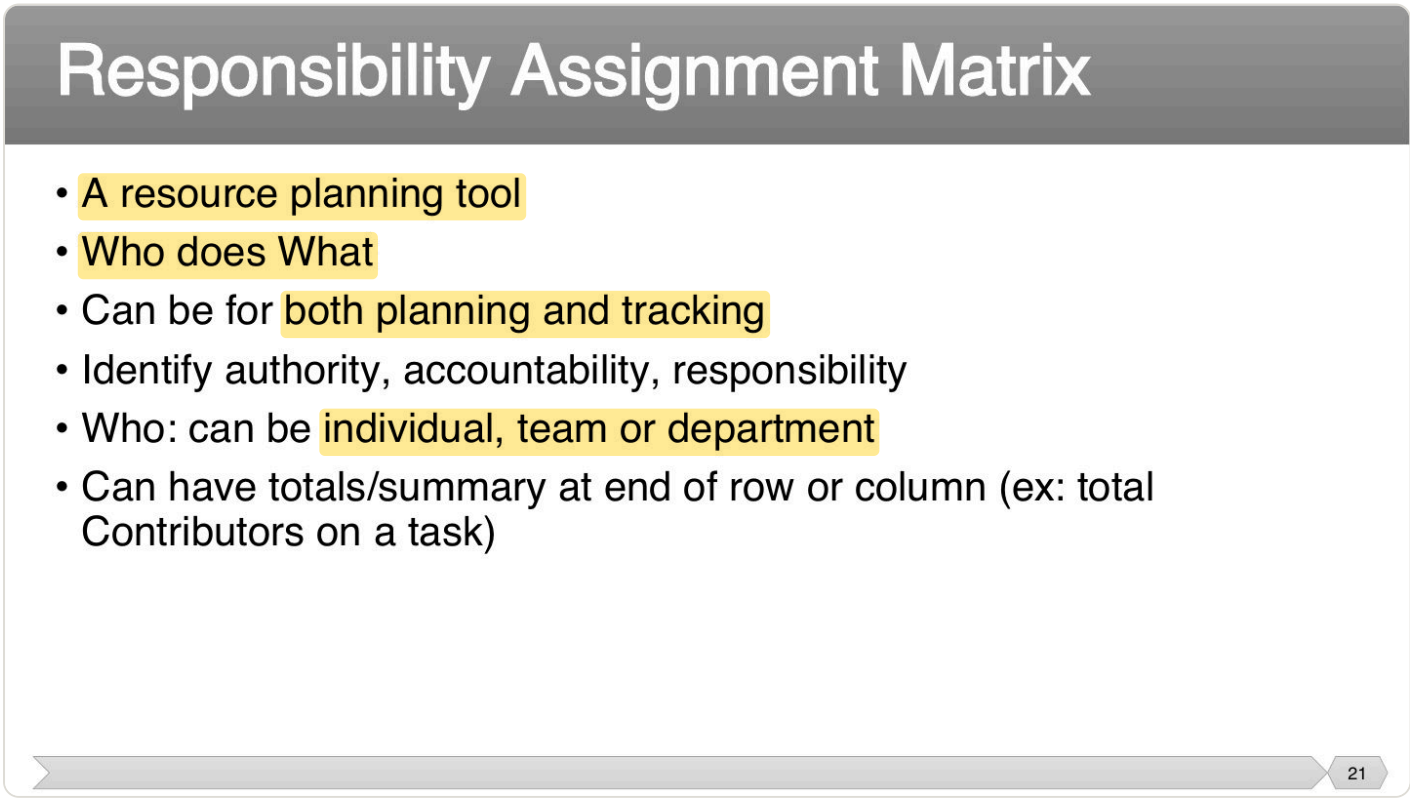
A Consulted

B Informed

C Responsible

D **Accountable**

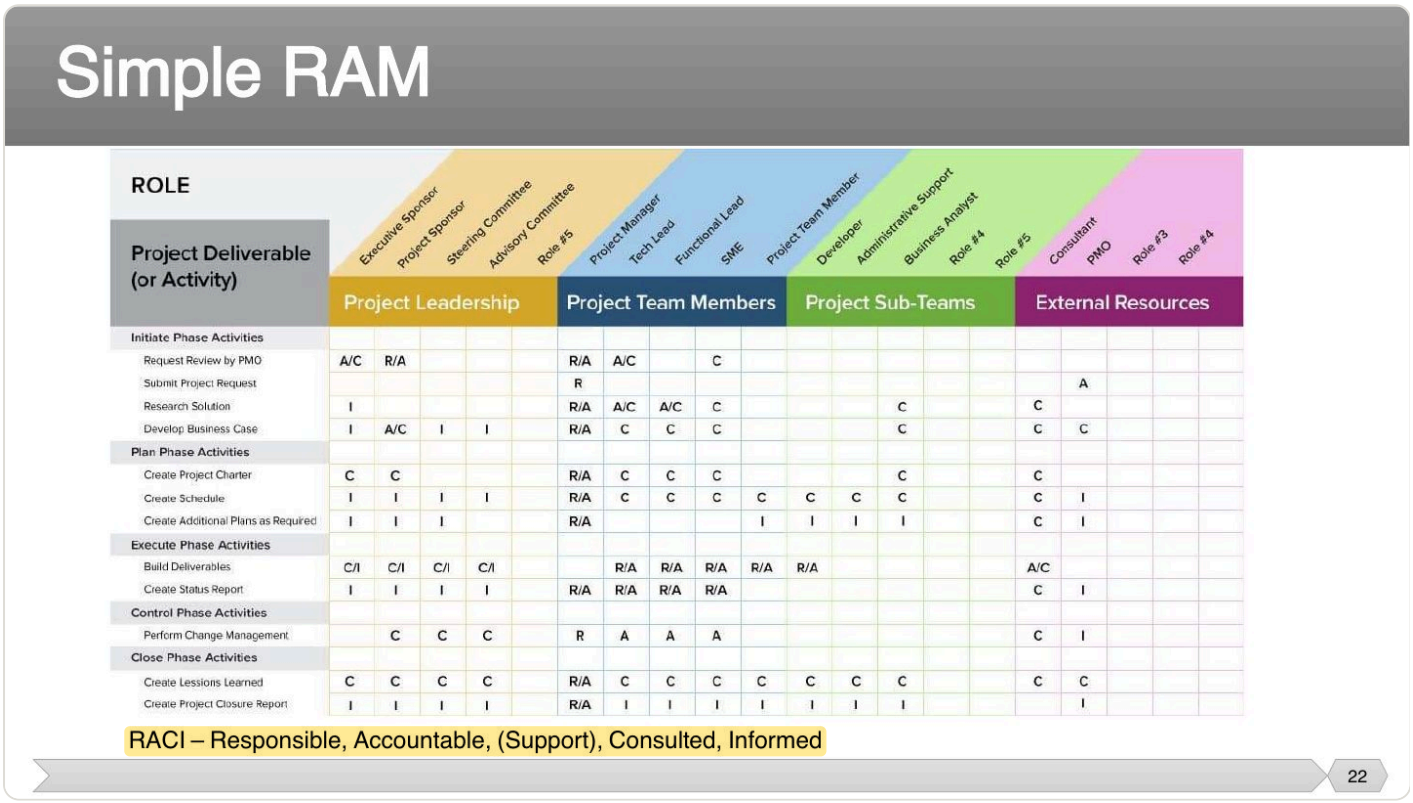
SLIDE 21 / 77 **Responsibility Assignment Matrix**



NOTES

1. A **Responsibility Assignment Matrix (RAM)** is a resource planning tool showing **who does what**; RACI is one kind of RAM.
2. It is used for both **planning** (assigning work) and **tracking** (checking each task has an owner), and makes authority, accountability and responsibility visible.
3. The "who" can be an **individual, team or department**, and rows or columns can have totals (e.g. the number of contributors on a task).

SLIDE 22 / 77 Simple RAM



NOTES

1. A real RAM: project deliverables grouped by phase (initiate, plan, execute, control, close) down the side, and roles grouped as project leadership, team members, sub-teams and external resources across the top. Each cell holds a RACI letter, so you can read across a row to see who is involved in a deliverable, or down a column to see one role's workload.

 Skills Matrix

Skills Matrix

- Another resource planning tool
- Resources on one axis, skills on other
- Skills can be high level or very specific
- Cells can be X's or numeric (ex: level, # yrs.)



23

NOTES

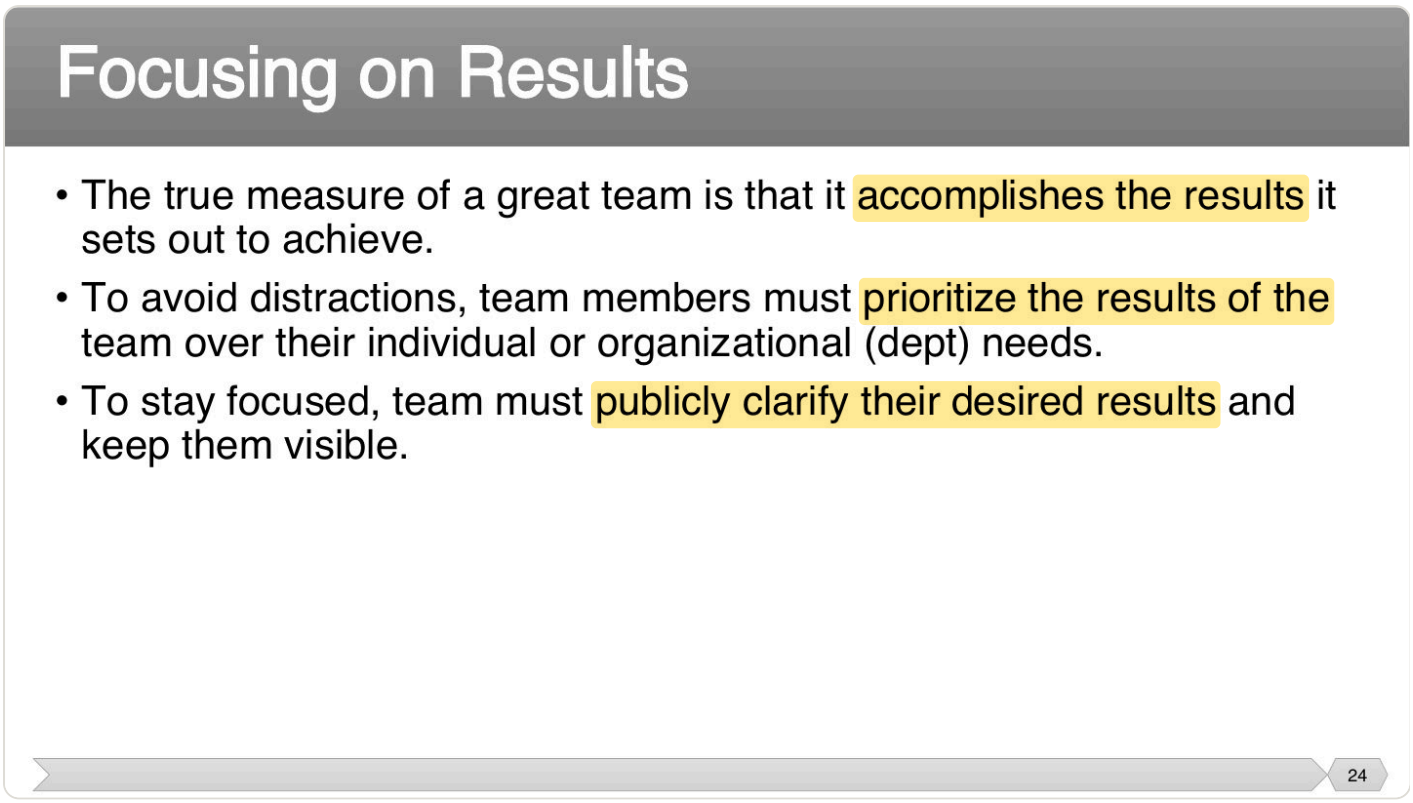
1. A **skills matrix** is another resource planning tool: **people on one axis, skills on the other**.
2. Skills can be high level or very specific (PHP, MySQL, React); cells can be X's or **numeric** levels (the legend: expert, practitioner, novice). Use it to assign tasks to the right people and to spot gaps, e.g. only one person is an expert in a key technology, which is a key-person risk.

EXAM ANGLE

Past exam · Final 242 · Q2.3 Q1

You are responsible for managing a diverse team with varying availability and skill sets. Describe two strategies you would use to allocate your team's resources efficiently across tasks. How would you adjust your resource plan if one hardware specialist becomes unavailable for 4 weeks?
Answer: 1) Build a skills matrix and assign tasks to the people whose skills match, recording owners in a RAM/RACI. 2) Check availability per period and level the workload so nobody is overallocated, using float on non-critical tasks. If the hardware specialist is out for 4 weeks: move their non-critical tasks later within their float, reassign what others can do (using the skills matrix to find the next-best person), bring in a contractor or borrow a specialist for critical work, and update the schedule and inform stakeholders if the critical path moves.

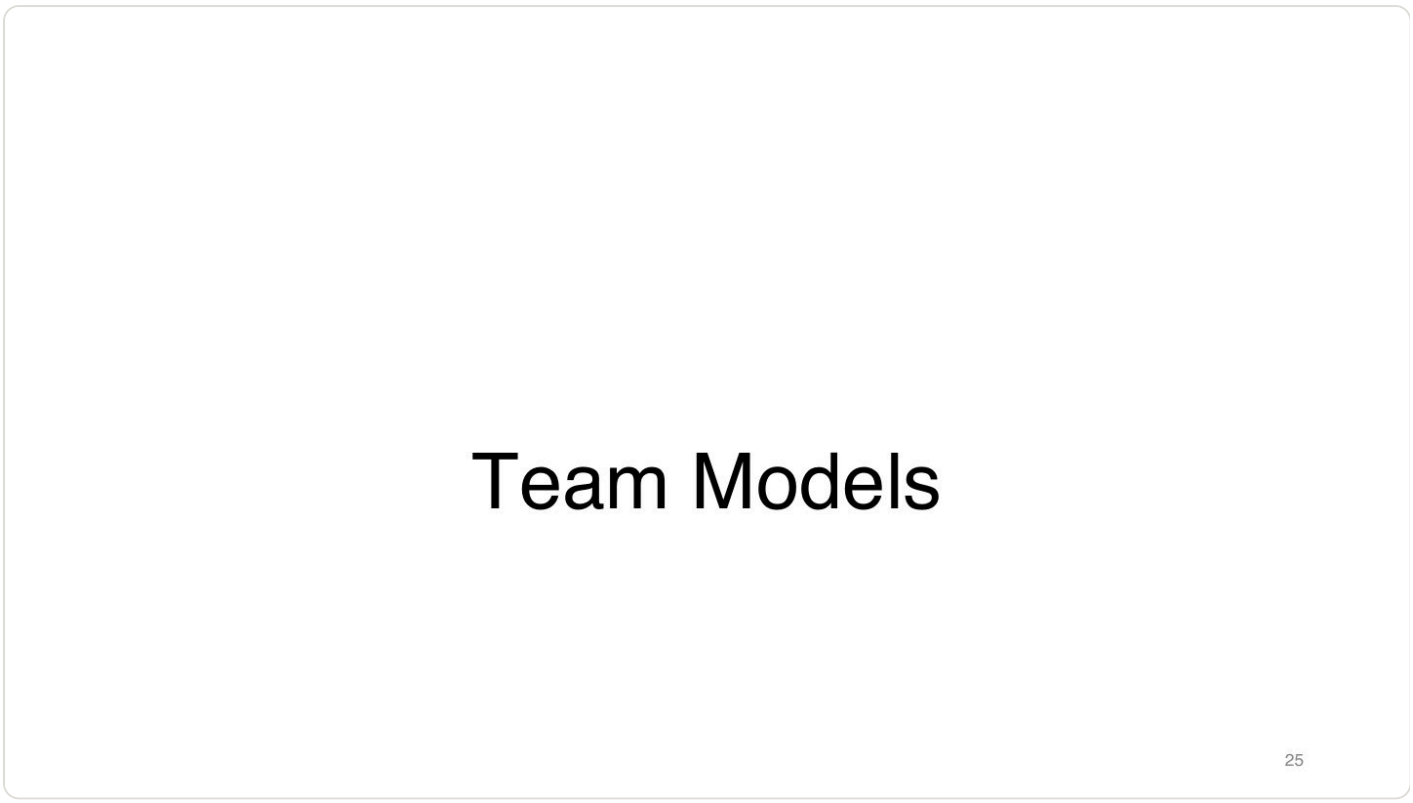
 Focusing on Results



NOTES

1. The top of the pyramid. The true measure of a great team is that it **accomplishes the results** it set out to achieve.
2. Members must **prioritize team results** over individual or departmental needs (status, credit, their department's agenda).
3. To stay focused, the team **publicly clarifies its desired results** and keeps them visible, e.g. sprint goals on the board.

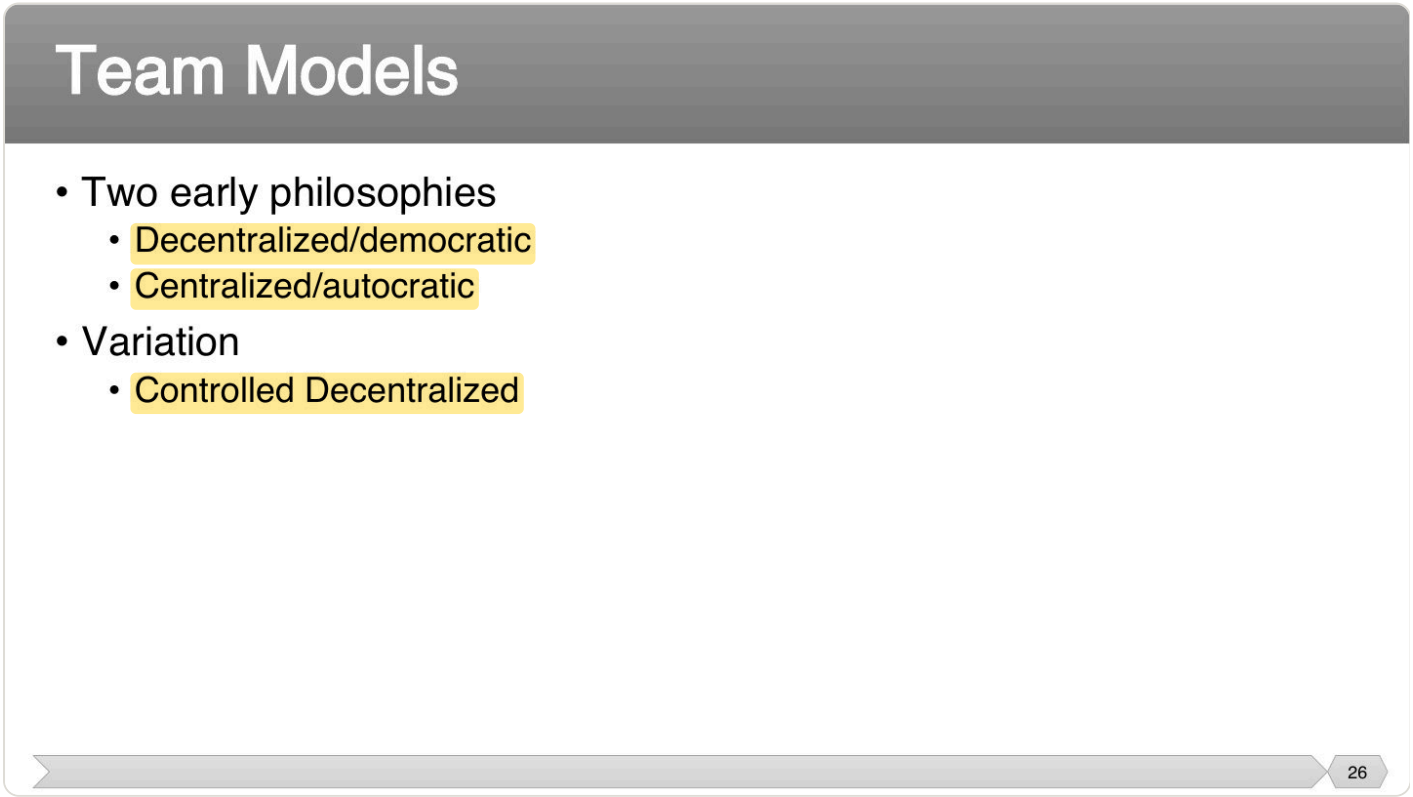
SLIDE 25 / 77 **Slide 25**



NOTES

1 Section divider: **Team Models**, the ways a software team can be organized.

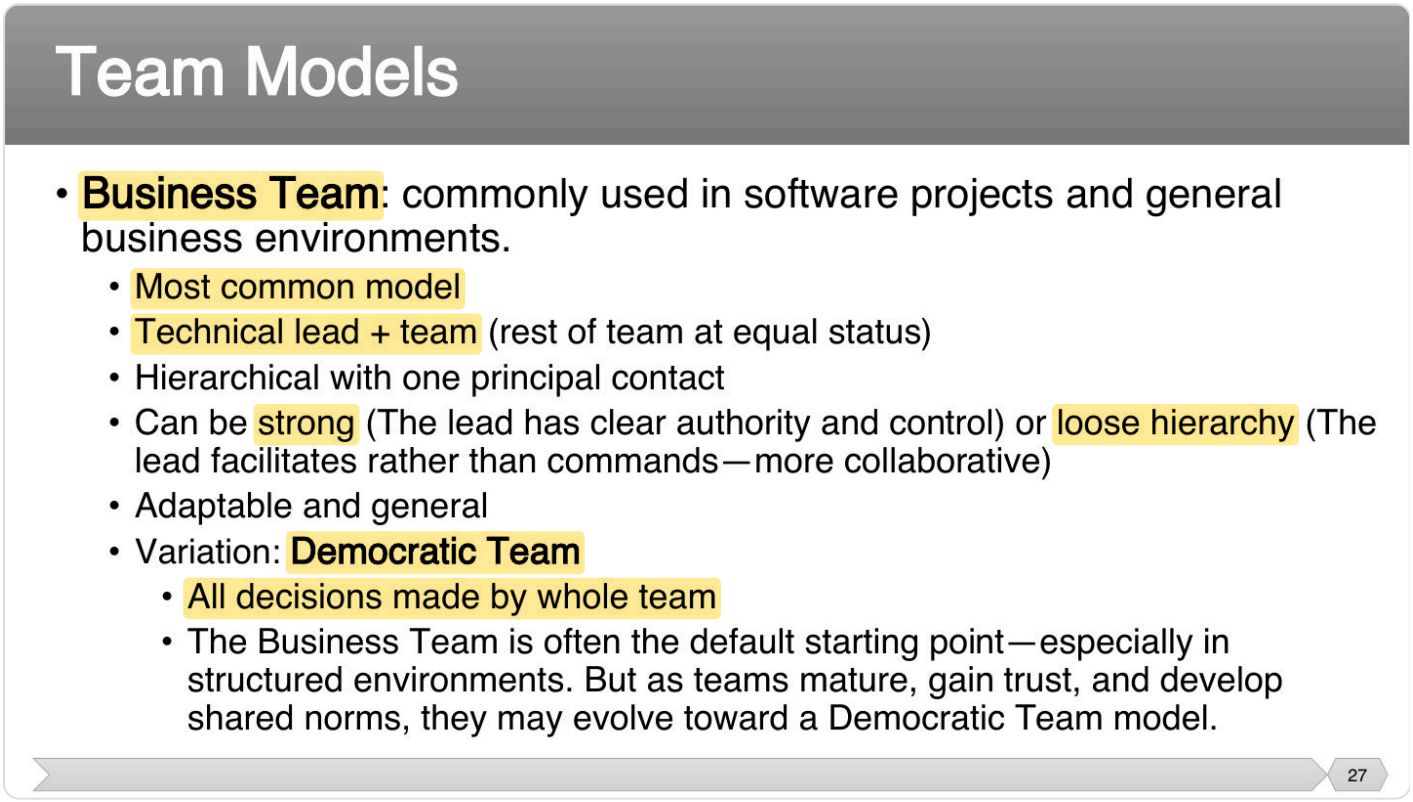
SLIDE 26 / 77 **Team Models**



NOTES

1. The two early philosophies: **decentralized/democratic** (the team decides together) and **centralized/autocratic** (one person decides).
2. The variation in between, **controlled decentralized**, has a leader who coordinates while sub-groups decide within their own areas.

SLIDE 27 / 77 **Team Models**



NOTES

1. The **business team** is the **most common** model in software and general business.
2. A **technical lead plus the team** (the others at equal status), hierarchical with one principal contact. It can be a **strong** hierarchy (the lead has clear authority) or a **loose** one (the lead facilitates rather than commands). It is adaptable and general.
3. The variation, the **democratic team**: all decisions are made by the whole team. Teams often start as a business team and evolve toward democratic as they mature, gain trust and develop shared norms.

TIPS & TRICKS

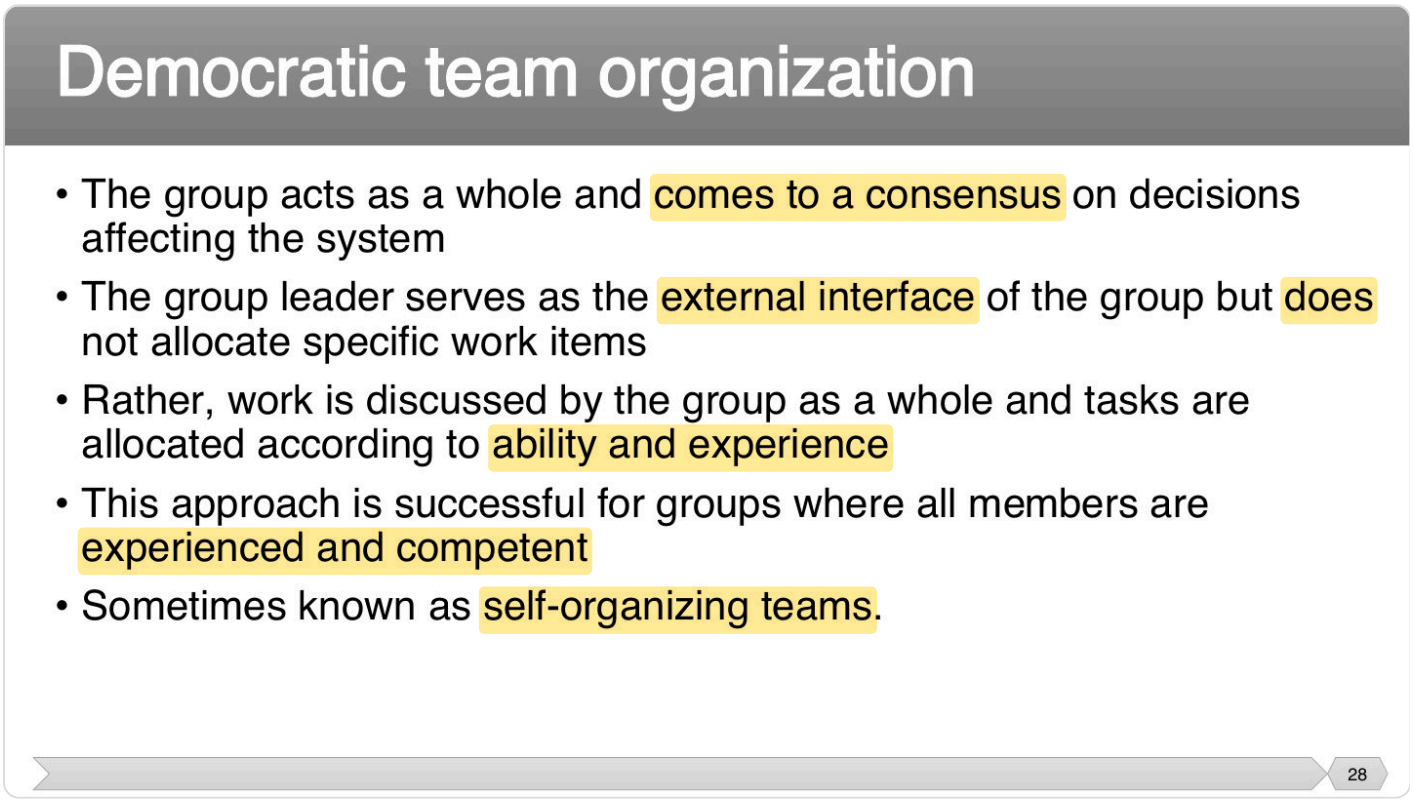
- Team models, in slide order: **B**usy **D**evelopers **C**ode **S**uper **S**oftware. Business, Democratic, Chief-programmer, Skunk works, SWAT.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q4 (also V2 Short Q4) Q1

What's the difference between a Business Team and a Democratic Team?
Answer: A business team has a technical lead who is the principal contact and makes or facilitates decisions, with the other members at equal status under the lead (a strong or loose hierarchy); it is the most common model. A democratic team acts as a group: decisions are made by consensus, the leader is only the external interface and does not allocate work, tasks are allocated by ability and experience, and it works best with experienced, competent members. Teams often move from business to democratic as trust and norms develop.

SLIDE 28 / 77 **Democratic team organization**



NOTES

1. In a **democratic team** the group acts as a whole and **comes to a consensus** on decisions affecting the system.
2. The leader is only the **external interface** and does **not** allocate specific work items.
3. Work is discussed by the whole group and tasks are allocated by **ability and experience**.
4. It succeeds only when **all members are experienced and competent**; such teams are also called **self-organizing teams** (as in Scrum).

SLIDE 29 / 77 **Agile programming groups**

Agile programming groups

- Agile programming groups are variants of democratic organization
- In Agile programming groups, some 'management' decisions are delegated to group members
- In XP, Programmers work in pairs and take a collective responsibility for code that is developed

NOTES

1. **Agile groups** are variants of democratic organization: some management decisions (estimating, task assignment) are delegated to the group members.
2. In **XP**, programmers work in pairs and take **collective responsibility** for the code: anyone can change any part, and nobody "owns" a module.

SLIDE 30 / 77 **Team Models**

Team Models

Chief-Programmer Team a.k.a. 'surgical team': From IBM in 70's*:* Puts a superstar at the top: Just as a surgeon performs the critical operation while being supported by anesthetists, nurses, and technicians, the **chief programmer** writes the core code while being supported by a carefully selected team of specialists.

- Backup Programmer/co-pilot (Equally skilled—ready to step in if needed, often reviews or shadows the chief.)
- Program Clerk (Manages documentation, version control, and technical records.)
- Administrator (Handles administrative tasks, freeing technical staff to focus on development.)
- Tool-smith (Builds and maintains development tools and environments.)
- "Language lawyer" (Deep expert in the programming language(s) used—consulted for tricky syntax or semantics.)

- Issues: Difficult to achieve, Ego issues: superstar and/or team
- Can be appropriate for creative projects or tactical execution

NOTES

1. The **chief-programmer (surgical) team**, from IBM in the 1970s: a superstar writes the core code, supported by specialists, like a surgeon supported by anesthetists and nurses.
2. The **backup programmer (co-pilot)** is equally skilled, shadows and reviews the chief, and can step in if needed.
3. The **program clerk** manages documentation and version control; the **administrator** handles admin work; the **tool-smith** builds and maintains tools and environments; the **language lawyer** is the deep expert in the programming language.
4. Issues: it is **difficult to achieve** (superstars are rare) and has **ego problems** on both sides. It can suit creative projects or tactical execution.

TIPS & TRICKS

- **BAT CLiP**: Backup programmer, Administrator, Tool-smith, Chief programmer, Language lawyer, Program clerk.

EXAM ANGLE

Prediction Q1

In a chief-programmer team, who is equally skilled and ready to step in for the chief?

A The language lawyer

B The tool-smith

C The program clerk

D **The backup programmer (co-pilot)**

 Chief programmer teams

Chief programmer teams

- Consist of a kernel of specialists helped by others added to the project as required
- The motivation behind their development is the wide difference in ability in different programmers
- Chief programmer teams provide a supporting environment for very able programmers to be responsible for most of the system development

NOTES

1. A chief-programmer team is a **kernel of specialists**, with others added to the project as required.
2. The motivation: the **wide difference in ability** between programmers. If the best are many times more productive, build the team around them.
3. It gives **very able programmers** a supporting environment so they can be responsible for most of the system.

 Problems

Problems

- This chief programmer approach, in different forms, has undoubtedly been successful
- However, it suffers from a number of problems
  - Talented designers and programmers are hard to find. Without exceptional people in these roles, the approach will fail
  - Other group members may resent the chief programmer taking the credit for success so may deliberately undermine his/her role.
  - High project risk as the project will fail if both the chief and deputy programmer are unavailable
  - Organizational structures and grades may be unable to accommodate this type of group

NOTES

1. Problems with the chief-programmer approach. Talented designers and programmers are **hard to find**; without exceptional people it fails.
2. Other members may **resent the chief taking the credit** and deliberately undermine them.
3. **High project risk**: the project fails if both the chief and the deputy are unavailable (a key-person risk).
4. The organization's **structures and pay grades** may not accommodate a team where one person is far above the rest.

EXAM ANGLE

Prediction Q1

A chief-programmer team carries high project risk because the project fails if both the chief and the deputy are unavailable.

- T **True**
- F False

 Team Models

Team Models

- Skunk works Team
  - Put a bunch of talented, creative developers away from the mother ship
    - Off-site literally or figuratively
  - Pro: Creates high ownership & buy-in
  - Con: Little visibility into team progress
  - Applicable: exploratory projects needing creativity
    - Not on well-defined or narrow problem
- SWAT Team
  - Highly skilled team
  - Skills tightly match goal
  - Members often work together
  - Ex: security swat team, Oracle performance team

NOTES

1. **Skunk works**: a group of talented, creative developers placed away from the “mother ship”, literally or figuratively. Pro: **high ownership and buy-in**.
2. Con: **little visibility** into their progress. Use it for **exploratory projects** needing creativity, not for well-defined or narrow problems.
3. **SWAT team**: highly skilled people whose **skills tightly match the goal**, who often work together repeatedly (e.g. a security SWAT team, an Oracle performance team).

EXAM ANGLE

Prediction Q1

A team of database experts is brought in to fix Oracle performance problems. This is a:

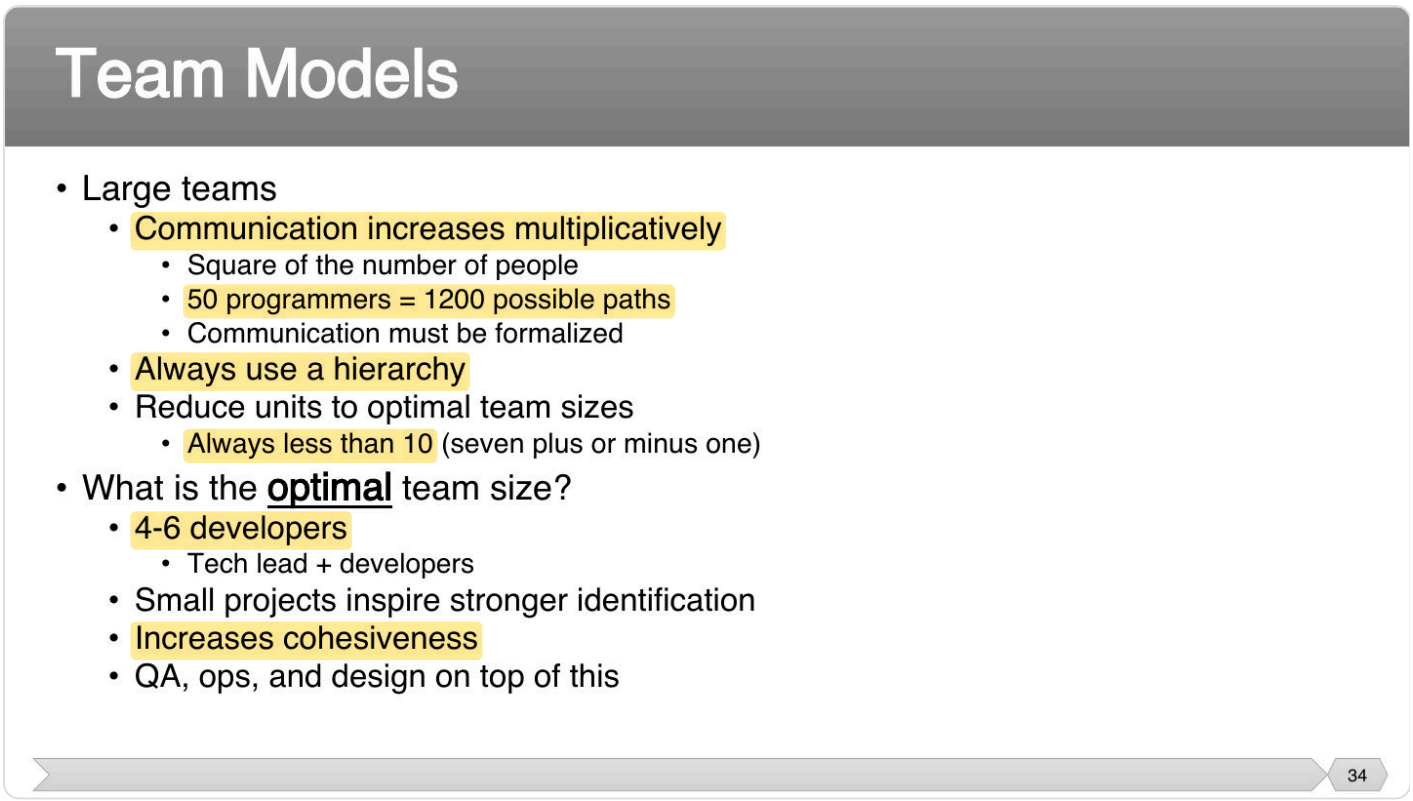
- **A SWAT team**
- B Skunk works team
- C Chief-programmer team
- D Democratic team

Prediction Q2

A company sends a few creative developers off-site to explore a new product idea with little reporting. This is a:

- A SWAT team
- B Business team
- C Chief-programmer team
- **D Skunk works team**

 Team Models



NOTES

1. In **large teams** communication paths grow with roughly the square of the team size. The exact formula is **n(n – 1) / 2**: 50 people → 1,225 paths (the slide rounds to 1,200); 7 people → 21.
2. So large teams must formalize communication, **always use a hierarchy**, and break into units of optimal size: **always fewer than 10** (seven plus or minus one).
3. The **optimal team size** is **4–6 developers** plus a tech lead, with QA, operations and design on top. Small teams inspire stronger identification and **cohesiveness**.

EXAM ANGLE

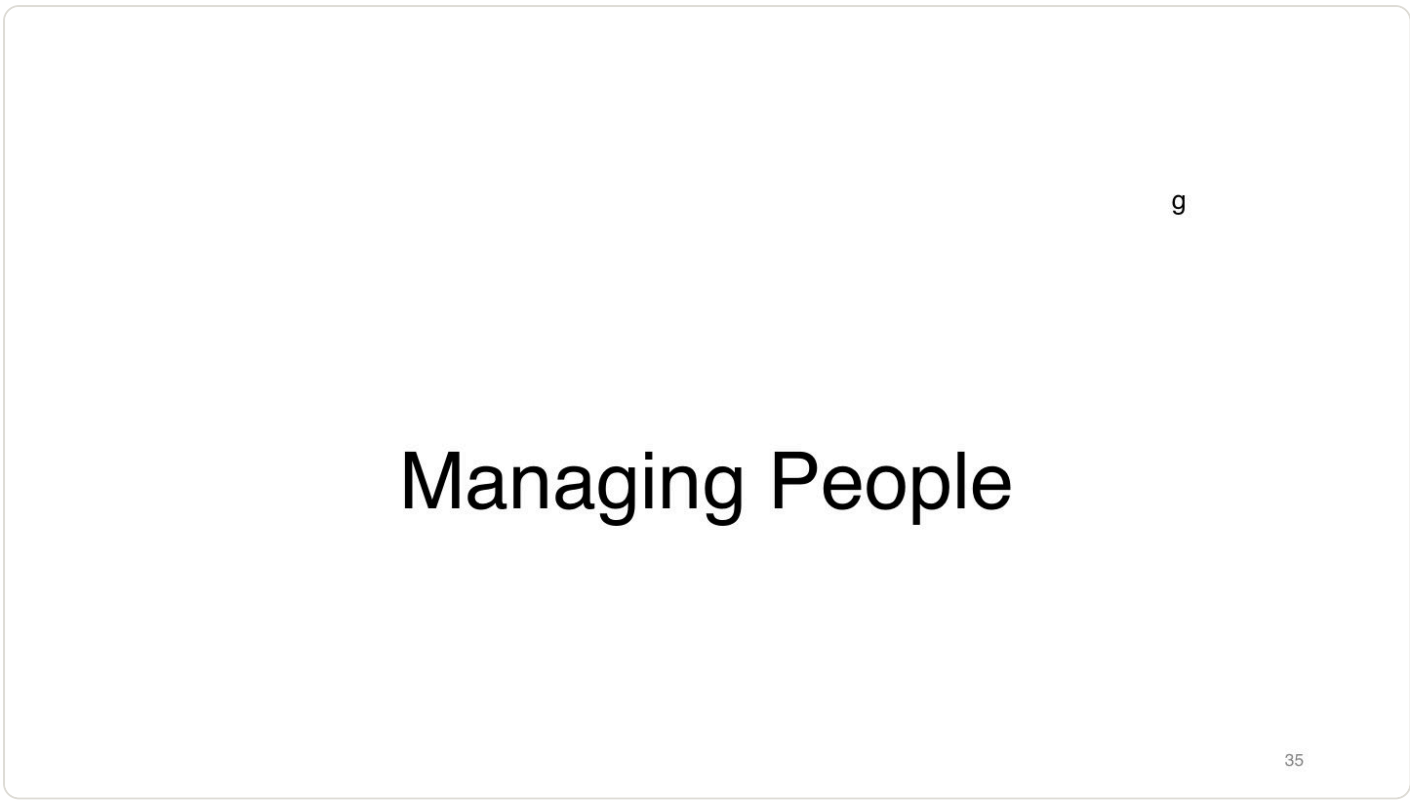
Past exam · Final 242 · Q3.2 Q1

Calculate the number of communication channels for a project team of 7 members.
Answer: n(n – 1) / 2 = 7 × 6 / 2 = 21 channels.

Prediction Q2

Your team grows from 5 to 10 people. By how much do the communication channels increase?
Answer: From 5 × 4 / 2 = 10 to 10 × 9 / 2 = 45 channels: the team doubled but channels grew 4.5 times. That is why large teams need hierarchy and should be split into sub-teams under 10.

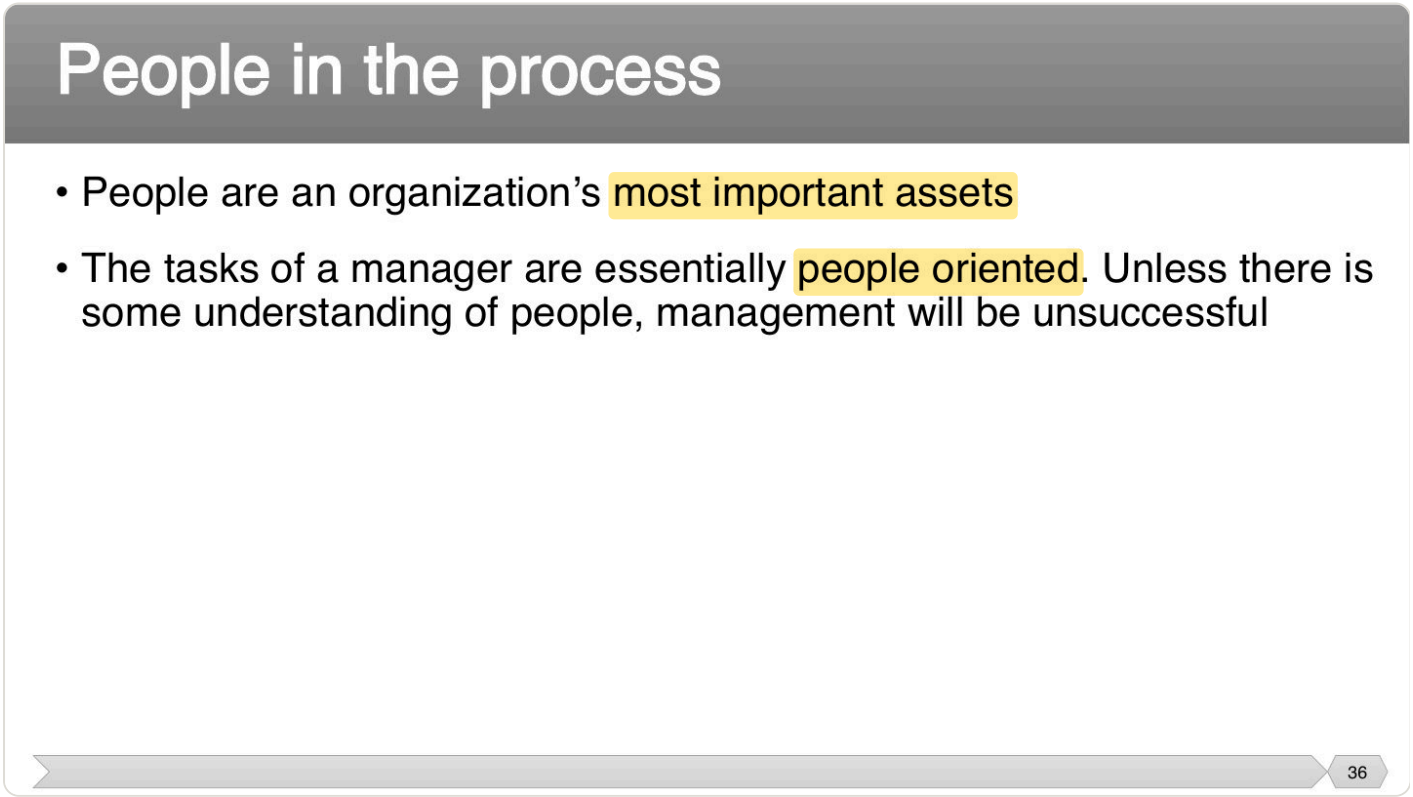
SLIDE 35 / 77 **Managing People**



NOTES

1 Section divider: **Managing People**.

SLIDE 36 / 77 **People in the process**

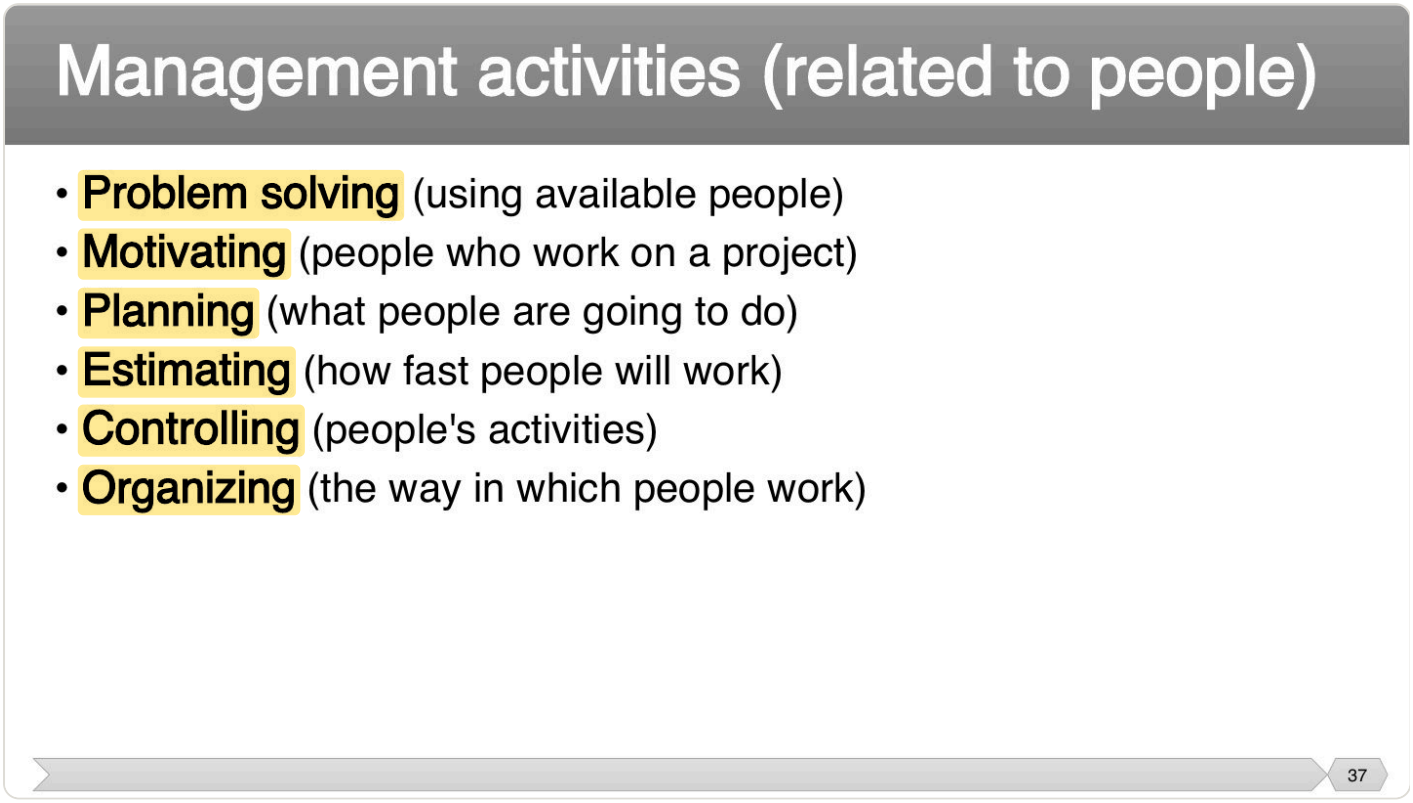


NOTES

1 People are an organization's **most important assets**.

2. A manager's tasks are essentially **people-oriented**; without understanding people, management fails however good the technical plan is.

SLIDE 37 / 77 **Management activities (related to people)**



NOTES

1. Every management activity is about people: **problem solving** with the people available and **motivating** those on the project...
2. ...**planning** what people will do and **estimating** how fast they will work...
3. ...**controlling** their activities and **organizing** the way they work.

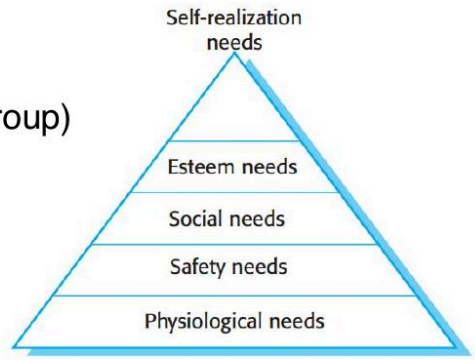
TIPS & TRICKS

- Sentence, in slide order: **P**roject **M**anagers **P**lan, **E**stimate, **C**ontrol, **O**rganize. Problem solving, Motivating, Planning, Estimating, Controlling, Organizing.

 Motivation

Motivation

- An important role of a manager is to motivate the people working on a project (motivation is needs-based)
- Motivation is a complex issue but it there are different types of motivation based on
  - Basic needs (e.g. food, sleep, etc.)
  - Personal needs (e.g. respect, self-esteem)
  - Social needs (e.g. to be accepted as part of a group)
- Maslow's hierarchy of needs



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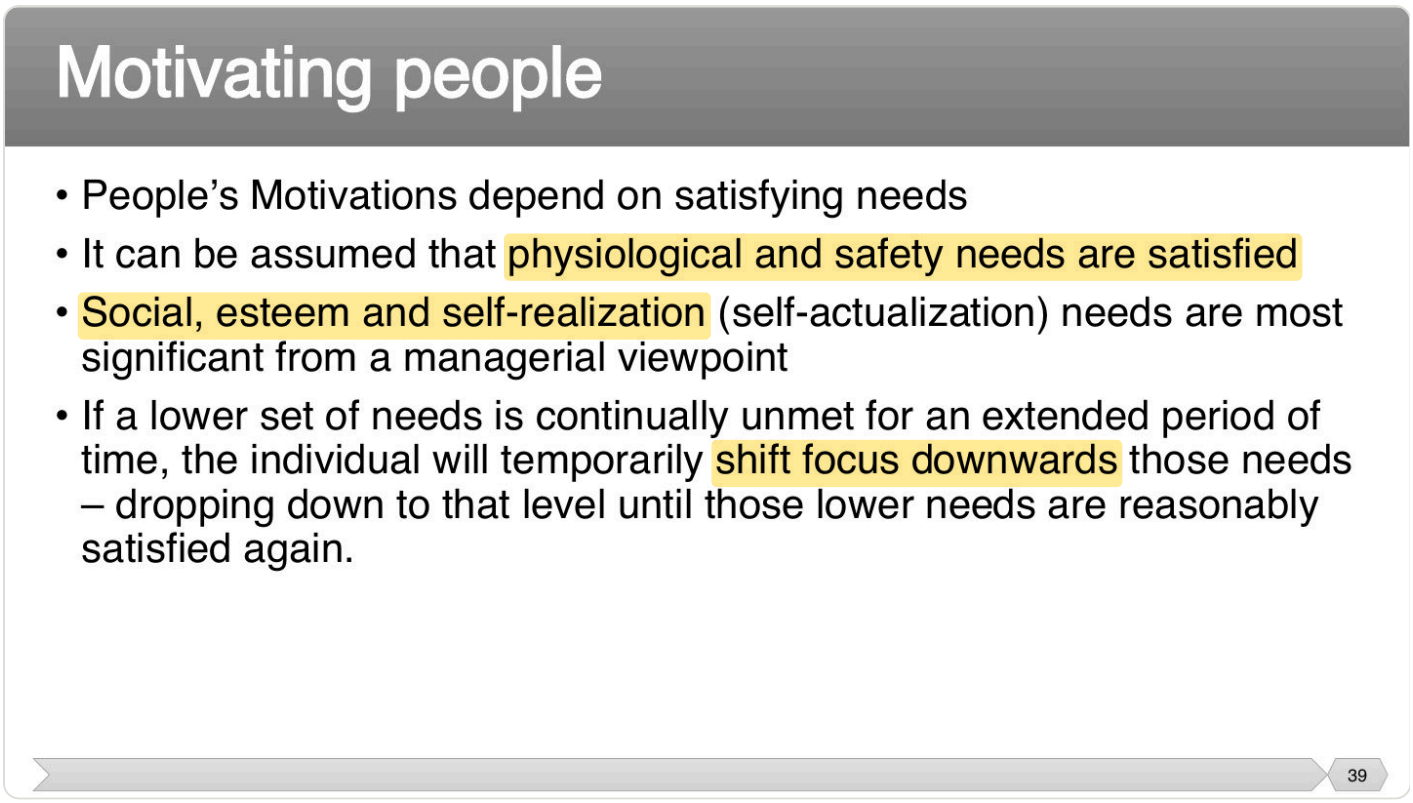
NOTES

1. A key role of the manager is to **motivate**, and motivation is **needs-based**: people act to satisfy needs.
2. Three kinds: **basic** (food, sleep), **personal** (respect, self-esteem) and **social** (being accepted as part of a group).
3. **Maslow's hierarchy**: physiological → safety → social → esteem → self-realization (self-actualization).

TIPS & TRICKS

- Maslow from the bottom up, where the order matters: **P**lease **S**tudents, **S**tudy **E**very **S**ession. Physiological, Safety, Social, Esteem, Self-realization.

 Motivating people



NOTES

1. At work you can assume **physiological and safety needs are satisfied** (salary, safe workplace).
2. So **social, esteem and self-realization** needs are the ones that matter for a manager.
3. But if a lower need stays unmet for a long time (e.g. job security during layoffs), people **shift focus downwards** to it until it is satisfied again, and higher motivators stop working.

 Need satisfaction

Need satisfaction

- Social
  - Provide communal facilities (shared spaces, break rooms, collaborative zones, etc.)
  - Allow informal communications (support casual conversations, team chats, non-work-related exchanges such as during lunch, etc.)
- Esteem
  - Recognition of achievements
  - Appropriate rewards (bonuses, promotions, certificates, symbolic gestures, etc.)
- Self-realization
  - Training – people want to learn more
  - Responsibility (empower team members to make decisions)

NOTES

1. How to satisfy each need. **Social**: communal facilities (shared spaces, break rooms) and informal communication (team chats, lunch).
2. **Esteem**: **recognition of achievements** and appropriate rewards (bonuses, promotions, certificates, symbolic gestures).
3. **Self-realization**: **training** (people want to learn) and **responsibility** (empower them to make decisions).

EXAM ANGLE

Past exam · Final 251 · Q5.3 — Q1

Your team has just completed a major milestone in the project development process. How would you celebrate this achievement collaboratively with your team members? How would you recognize individual contributions to the team's success? What actions would you take to ensure that team morale remains high as you move forward in the project?

Answer: Celebrate as a team (social need), e.g. a team lunch and an announcement of the milestone to the stakeholders. Recognize individuals specifically and publicly (esteem): name what each person contributed in the meeting, send thank-you notes to their managers, and give certificates or small rewards. Keep morale high (self-realization): hold a retrospective that captures what went well, set clear and achievable next goals, balance the workload, and offer training and more responsibility to those who want to grow.

Prediction — Q2

Sending a developer on a cloud-architecture course mainly satisfies which need?

A Safety

B **Self-realization**

C Physiological

D Social

 Personality types

Personality types

- The needs hierarchy is almost certainly an over-simplification
- Motivation should also take into account different personality types:
  - Task-oriented
    - The motivation for doing the work is the work itself
  - Self-oriented
    - The work is a means to an end which is the achievement of individual goals – e.g. to get rich, to buy a house, to travel etc.
  - Interaction-oriented
    - The principal motivation is the presence and actions of co-workers.
    - People go to work because they like to go to work

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NOTES

1. The needs hierarchy is an **over-simplification**; motivation also depends on **personality type**.
2. **Task-oriented**: the motivation is the work itself (they love solving the problem).
3. **Self-oriented**: the work is a means to personal goals (get rich, buy a house, travel).
4. **Interaction-oriented**: the main motivation is the presence and actions of co-workers; they come to work because they like being with the team.

TIPS & TRICKS

- **SIT**: Self-oriented, Interaction-oriented, Task-oriented.

EXAM ANGLE

Prediction Q1

A developer works mainly to save for a house. This personality type is:

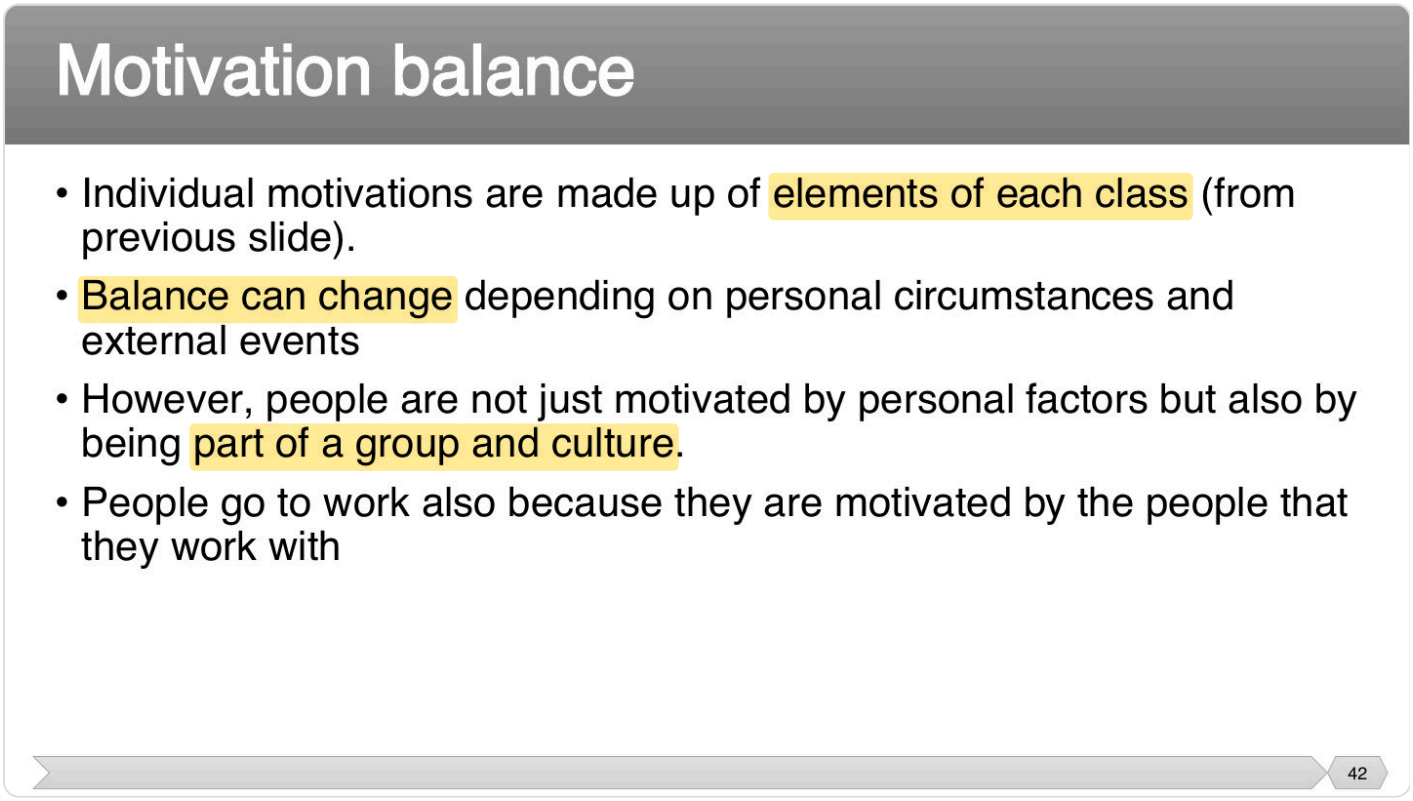
A Interaction-oriented

B Democratic

C Task-oriented

D **Self-oriented**

SLIDE 42 / 77 **Motivation balance**



NOTES

1. Real people have **elements of each class**, and the balance changes with personal circumstances (a new baby makes someone more self-oriented for a while).
2. People are also motivated by being **part of a group and culture**: they go to work partly for the people they work with.

 Group composition: group working

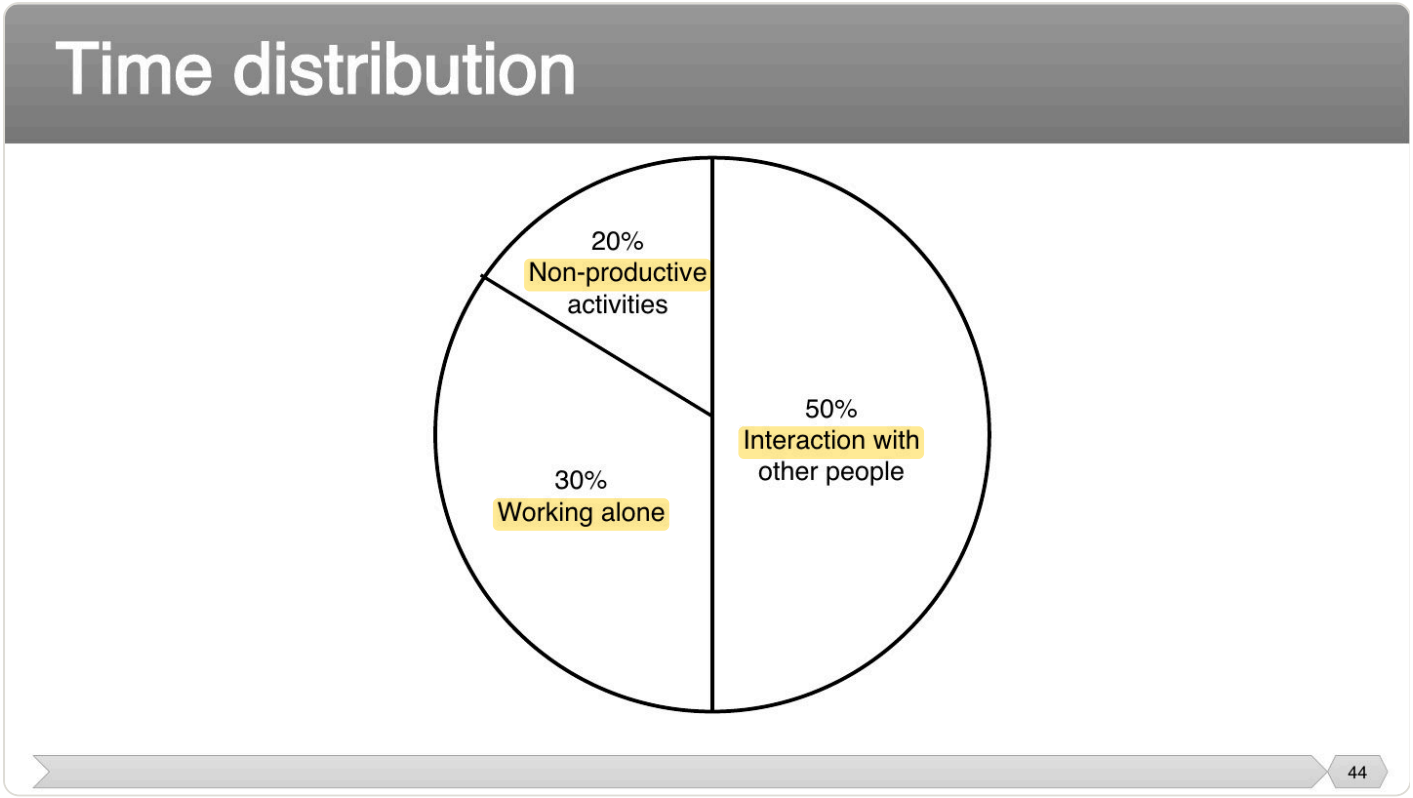
Group composition: group working

- Most software engineering is a group activity
  - The development schedule for most non-trivial software projects is such that they cannot be completed by one person working alone
- Group interaction is a key determinant of group performance
- Flexibility in group composition is limited
  - Managers must do the best they can with available people

NOTES

1. Most software engineering is a **group activity**; non-trivial schedules can't be met by one person.
2. **Group interaction** is a key determinant of performance, so how the group works together matters as much as individual skill.
3. Flexibility in **group composition is limited**: managers rarely pick an ideal team and must do their best with the people available.

 Time distribution



NOTES

1. About **50%** of an engineer's time is spent interacting with other people (meetings, reviews, discussions), so communication skills matter far more than people expect.
2. Only about **30%** is working alone, and **20%** goes on non-productive activities (admin, waiting, travel).

TIPS & TRICKS

- **WIN**: Working alone, Interaction with others, Non-productive.

EXAM ANGLE

Prediction Q1

According to the time-distribution chart, roughly how much of an engineer's time is spent interacting with other people?

A 80%

B 20%

C 30%

D **50%**

 Group composition

Group composition

- Group composed of members who share the same motivation can be problematic:
  - Task-oriented – everyone wants to do their own thing
  - Self-oriented – everyone wants to be the boss
  - Interaction-oriented – too much chatting, not enough work
- An effective group has a balance of all types
- Can be difficult to achieve because most engineers are task-oriented

NOTES

1. A group whose members all share the **same motivation** is a problem.
2. All **task**-oriented → everyone wants to do their own thing; all **self**-oriented → everyone wants to be the boss; all **interaction**-oriented → too much chatting, not enough work.
3. An effective group has a **balance of all types**, which is hard because **most engineers are task-oriented**. The interaction-oriented members are valuable because they keep communication going.

EXAM ANGLE

Prediction Q1

A team where “everyone wants to be the boss” is dominated by which personality type?

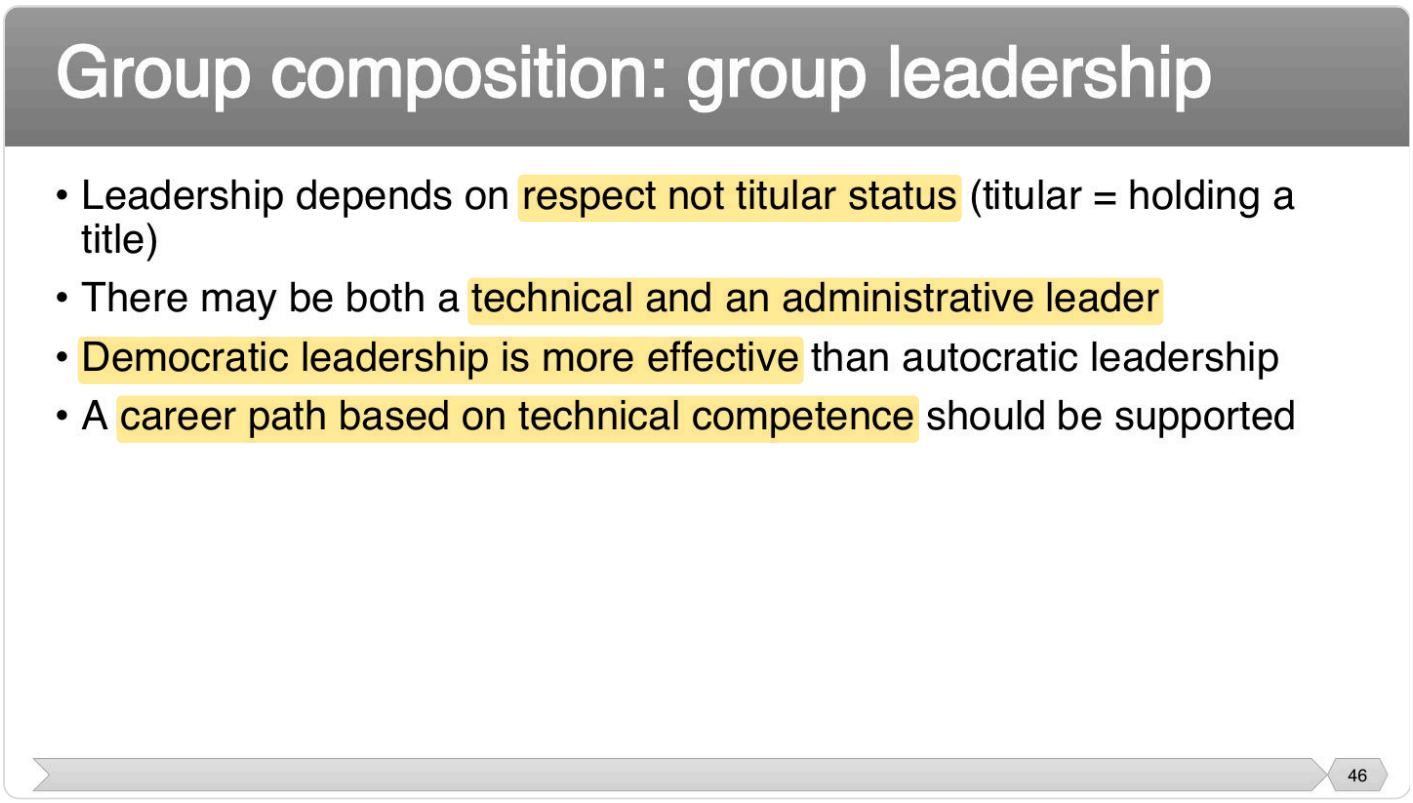
A Interaction-oriented

B Task-oriented

C **Self-oriented**

D Balanced

 Group composition: group leadership



NOTES

1. Leadership depends on **respect, not title**.
2. A group may have both a **technical leader** and an **administrative leader**.
3. **Democratic leadership** is more effective than autocratic in software groups.
4. Support a **technical career path** so good engineers don't have to become managers to progress.

Group cohesiveness

Group cohesiveness

- In a cohesive group, members consider the group to be more important than any individual in it
- Advantages of a cohesive group are:
  - Group quality standards can be developed
  - Group members work closely together so inhibitions caused by ignorance are reduced
  - Team members learn from each other and get to know each other's work
  - Egoless programming where members strive to improve each other's programs can be practiced

NOTES

1. In a **cohesive group** members consider the group **more important** than any individual in it.
2. Advantages: **group quality standards** can be developed and kept collectively; members work closely, so inhibitions caused by ignorance are reduced.
3. Members **learn from each other** and know each other's work, so anyone can cover for anyone.
4. **Egoless programming**: members try to improve each other's programs and don't treat code as personal property.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 10 (also V2 MCQ 9) Q1

In a cohesive team, which of the following is an advantage?

A Members avoid challenging each other's work to reduce conflict.

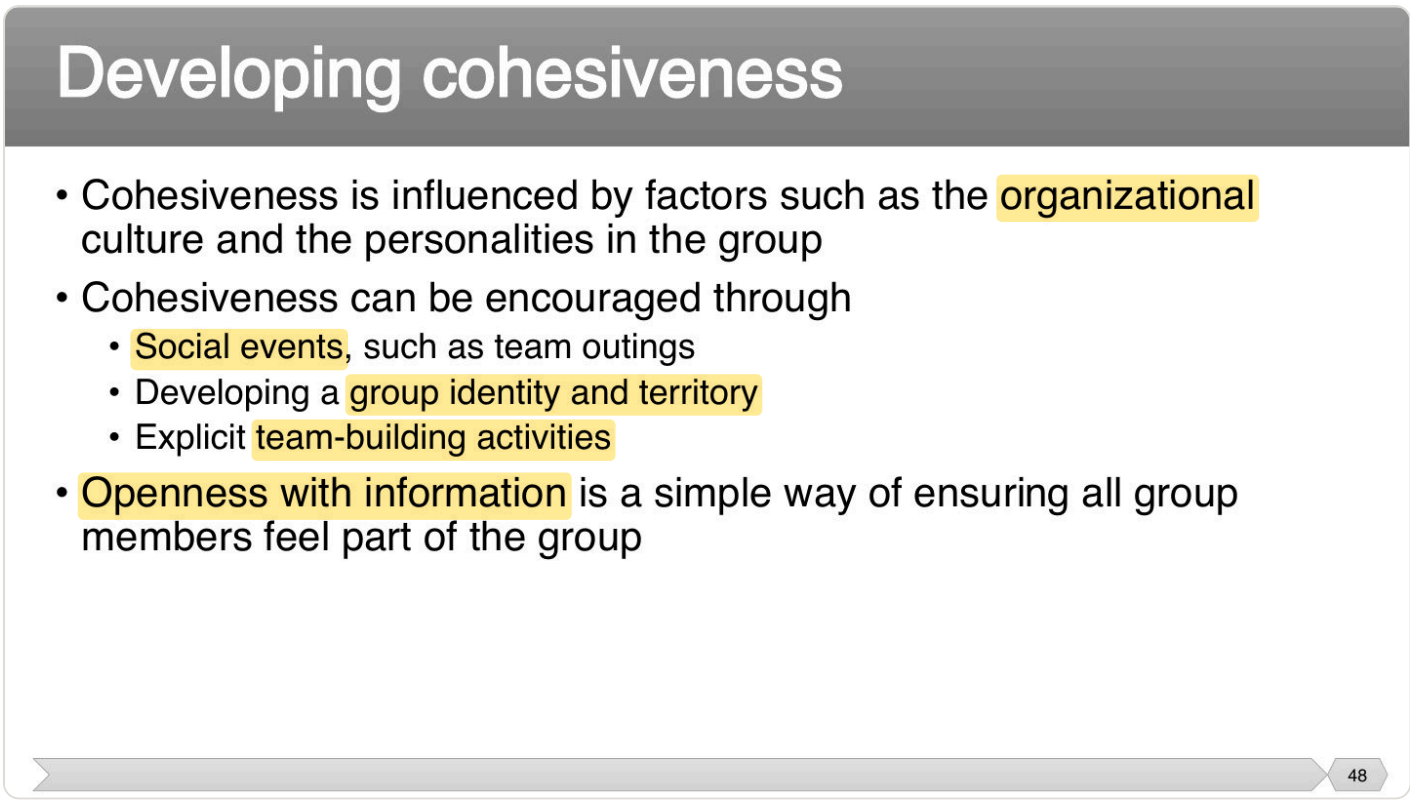
B **Group quality standards can be developed and maintained collectively.**

> Cohesive groups can set and keep shared quality standards, learn from each other and practise egoless programming. Avoiding challenge is groupthink, a risk of cohesion.

C Each member works independently without needing oversight.

D Decision-making is always faster due to consensus.

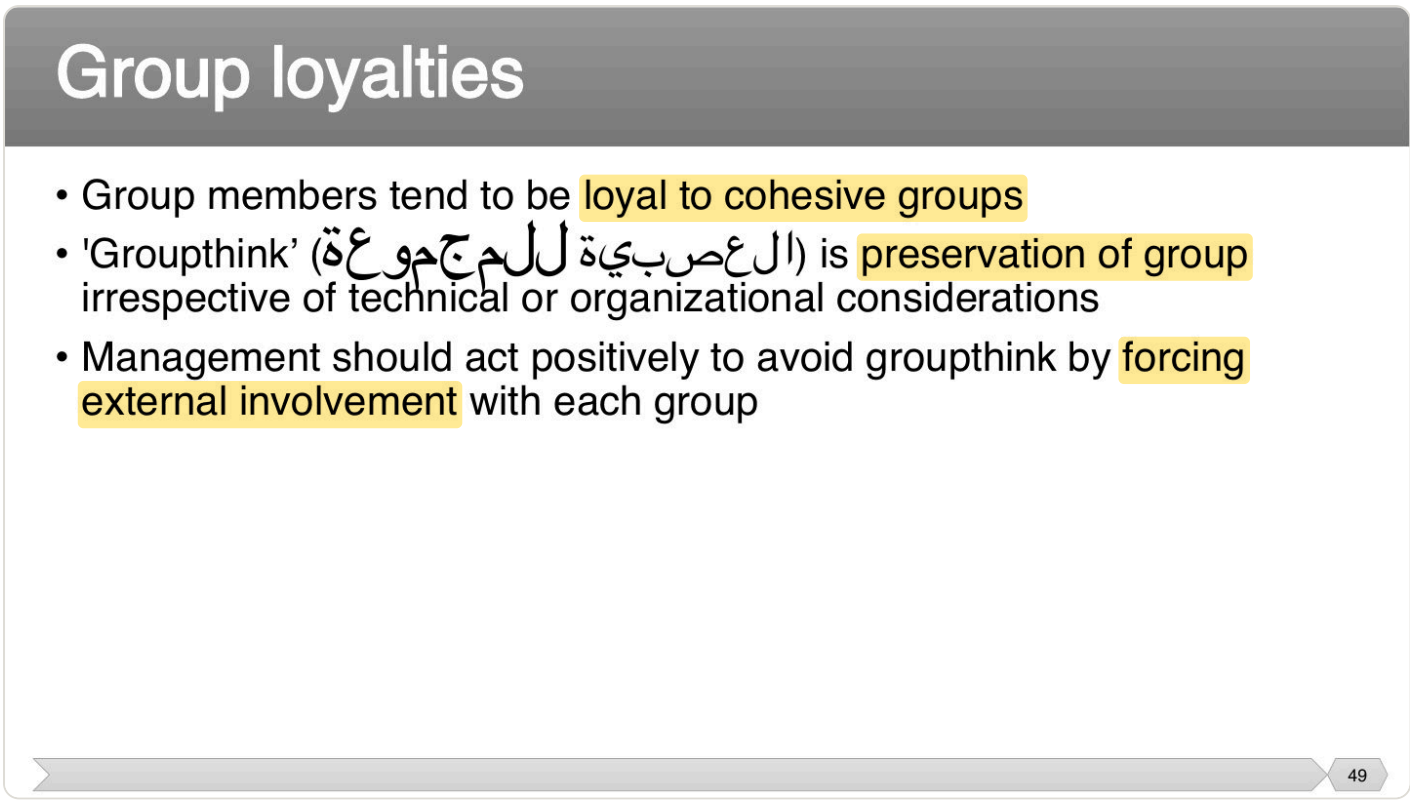
 Developing cohesiveness



NOTES

1. Cohesiveness is influenced by the **organizational culture** and the personalities in the group.
2. Encourage it through **social events** (team outings), a **group identity and territory** (a team name, its own space) and explicit **team-building**.
3. **Openness with information** is the simplest tool: when everyone knows what is going on, everyone feels part of the group.

 Group loyalties



NOTES

1. Members are **loyal to cohesive groups**, which is good until it isn't.
2. **Groupthink**: preserving the group regardless of technical or organizational considerations (defending a bad design because the group made it).
3. Management should prevent it by **forcing external involvement** with each group: outside reviewers, cross-team reviews, rotating members.

EXAM ANGLE

Prediction Q1

How should management counter groupthink?

A Reward unanimous agreement

B **Force external involvement with the group**

C Isolate the group further

D Remove all conflict

 Group communications

Group communications

- Good communications are essential for effective group working
- Information must be exchanged on the status of work, design decisions and changes to previous decisions
- Good communications also strengthens group cohesion as it promotes understanding

Notes:

- Status of group members
  - Higher status members tend to dominate conversations
- Personalities in groups
  - Too many people of the same personality type can be a problem
- Communication channels
  - Communications channelled though a central coordinator tend to be ineffective

NOTES

1. Good communication is essential: information on the **status of work, design decisions** and changes to them must be exchanged, and it **strengthens cohesion**.
2. Three things affect it. **Status**: higher-status members dominate conversations, so juniors' ideas get lost.
3. **Personalities**: too many of the same type is a problem (slide 45).
4. **Channels**: communication routed through a **central coordinator** tends to be ineffective, a bottleneck.

 Group organization

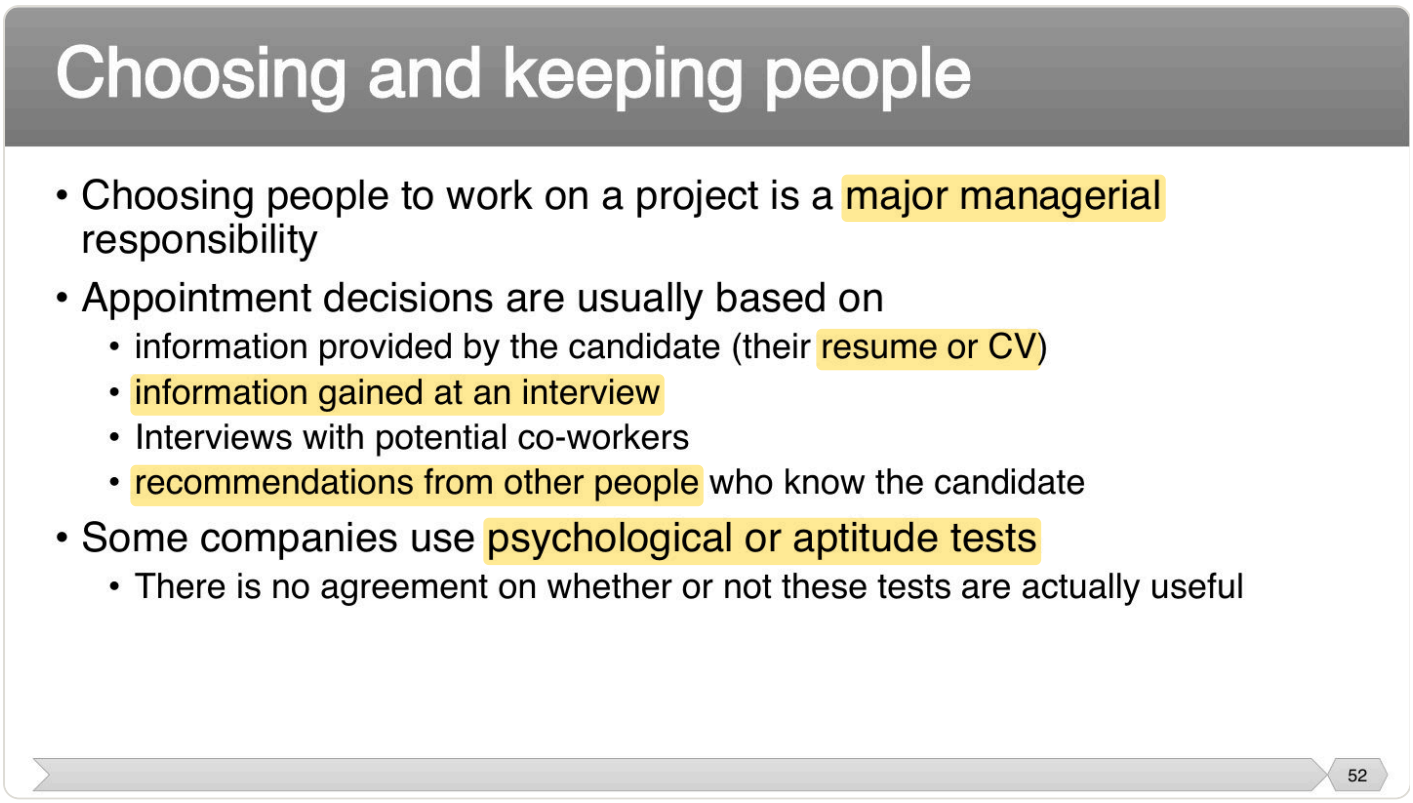
Group organization

- Software engineering group sizes should be relatively small (< 10 members) optimal is less than seven (see previous slides)
- Break big projects down into multiple smaller projects
- Small teams may be organized in an informal, democratic way
- Chief programmer teams try to make the most effective use of skills and experience

NOTES

1. Keep groups small (**< 10**, ideally under 7), and **break big projects down** into smaller ones.
2. Small teams can be organized in an **informal, democratic** way...
3. ...while **chief-programmer teams** try to make the most effective use of scarce skills and experience.

 Choosing and keeping people



NOTES

1. Choosing people is a **major managerial responsibility**.
2. Decisions are usually based on the candidate's **CV**, the **interview**, interviews with potential co-workers and **recommendations** from people who know them.
3. Some companies use **psychological or aptitude tests**, but there is no agreement on whether they are useful.

 Staff selection factors

Staff selection factors

Factor	Explanation
Application domain experience	For a project to develop a successful system, the developers must understand the application domain
Platform experience	May be significant if low-level programming is involved. Otherwise, not usually a critical attribute.
Programming language experience	Normally only significant for short duration projects where there is insufficient time to learn a new language.
Educational background	May provide an indicator of the basic fundamentals which the candidate should know and of their ability to learn. This factor becomes increasingly irrelevant as engineers gain experience across a range of projects.
Communication ability	Very important because of need for project staff to communicate orally and in writing with other engineers, managers, and customers.
Adaptability	Adaptability may be judged by looking at the different types of experience that candidates have had. This is an important attribute as it indicates an ability to learn.
Attitude	Project staff should have a positive attitude towards their work and should be willing to learn new skills. This is an important attribute but often very difficult to assess.
Personality	Again, an important attribute but difficult to assess. Candidates must be reasonably compatible with other team members. No particular type of personality is more or less suited to software engineering.

NOTES

1. **Application domain experience**: developers must understand the domain to build a successful system. One of the most important factors.
2. **Platform experience** matters only for low-level work; **programming language** experience matters only for short projects with no time to learn a new language. A common trap: it is not the most critical factor.
3. **Education** indicates fundamentals and ability to learn, and becomes less relevant with experience.
4. **Communication ability** is very important (staff talk to engineers, managers and customers); **adaptability** (judged from varied experience) shows ability to learn.
5. **Attitude** and **personality** are important but hard to assess. No personality type is better suited to software engineering; candidates must just fit with the team.

TIPS & TRICKS

- Sentence, in slide order: **A**ll **P**rogrammers **P**refer **E**asy **C**ode **A**nd **A**wesome **P**izza. Application domain, Platform, Programming language, Educational background, Communication ability, Adaptability, Attitude, Personality.

EXAM ANGLE

Prediction Q1

According to the table, programming language experience is significant mainly when:

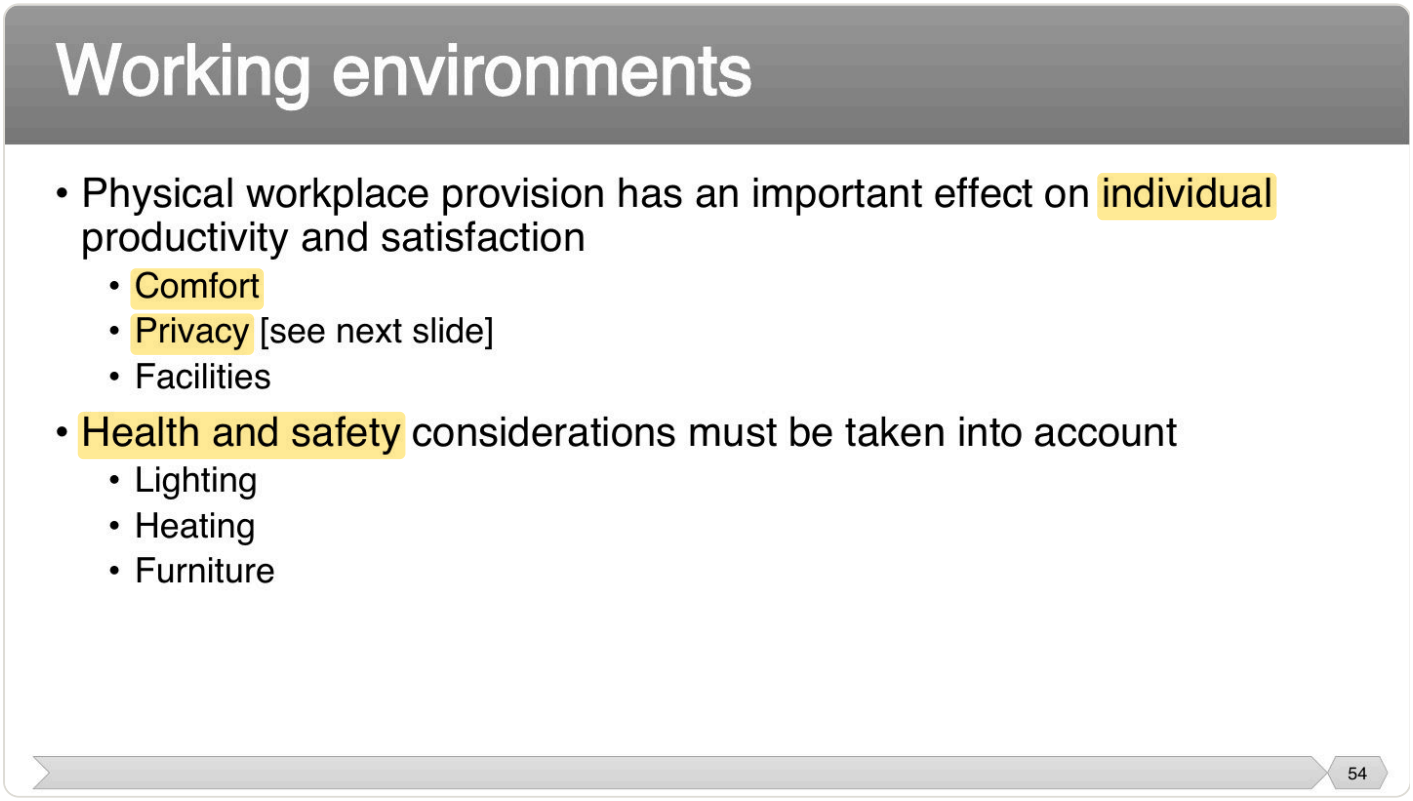
A The project is fully agile

B **The project is short with no time to learn a new language**

C The project is long

D The candidate has a postgraduate degree

SLIDE 54 / 77 **Working environments**



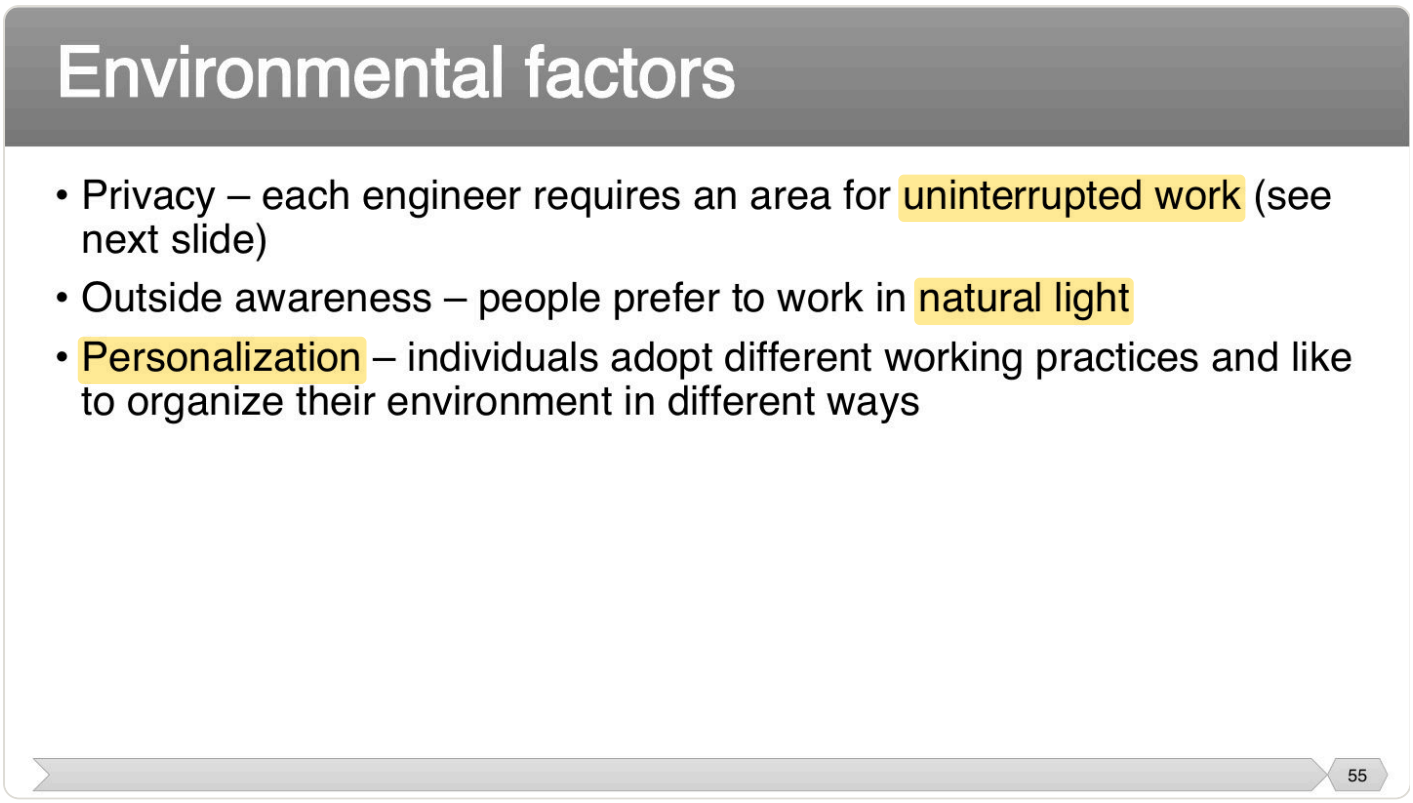
NOTES

1. The physical workplace affects **individual productivity and satisfaction**.
2. Provide **comfort**, **privacy** and facilities...
3. ...and take **health and safety** into account: lighting, heating, furniture.

TIPS & TRICKS

- **PaCiFy**: Privacy, Comfort, Facilities.

 Environmental factors



NOTES

1. **Privacy**: each engineer needs an area for **uninterrupted work**, because deep work is destroyed by interruptions.
2. **Outside awareness**: people prefer working in **natural light**.
3. **Personalization**: people organize their environment differently and like to do it their own way.

TIPS & TRICKS

- **POP**: Privacy, Outside awareness, Personalization.

SLIDE 56 / 77 **Workspace organization**

Workspace organization

- Workspaces should provide private spaces where people can work without interruption
  - Providing individual offices for staff has been shown to increase productivity
- However, teams working together also require spaces where formal and informal meetings can be held

NOTES

1. Provide private spaces to work **without interruption**; individual offices have been shown to **increase productivity**.
2. But teams also need shared spaces for **formal and informal meetings**. Balance focus and collaboration.

SLIDE 57 / 77 **Our perspective about the team members**

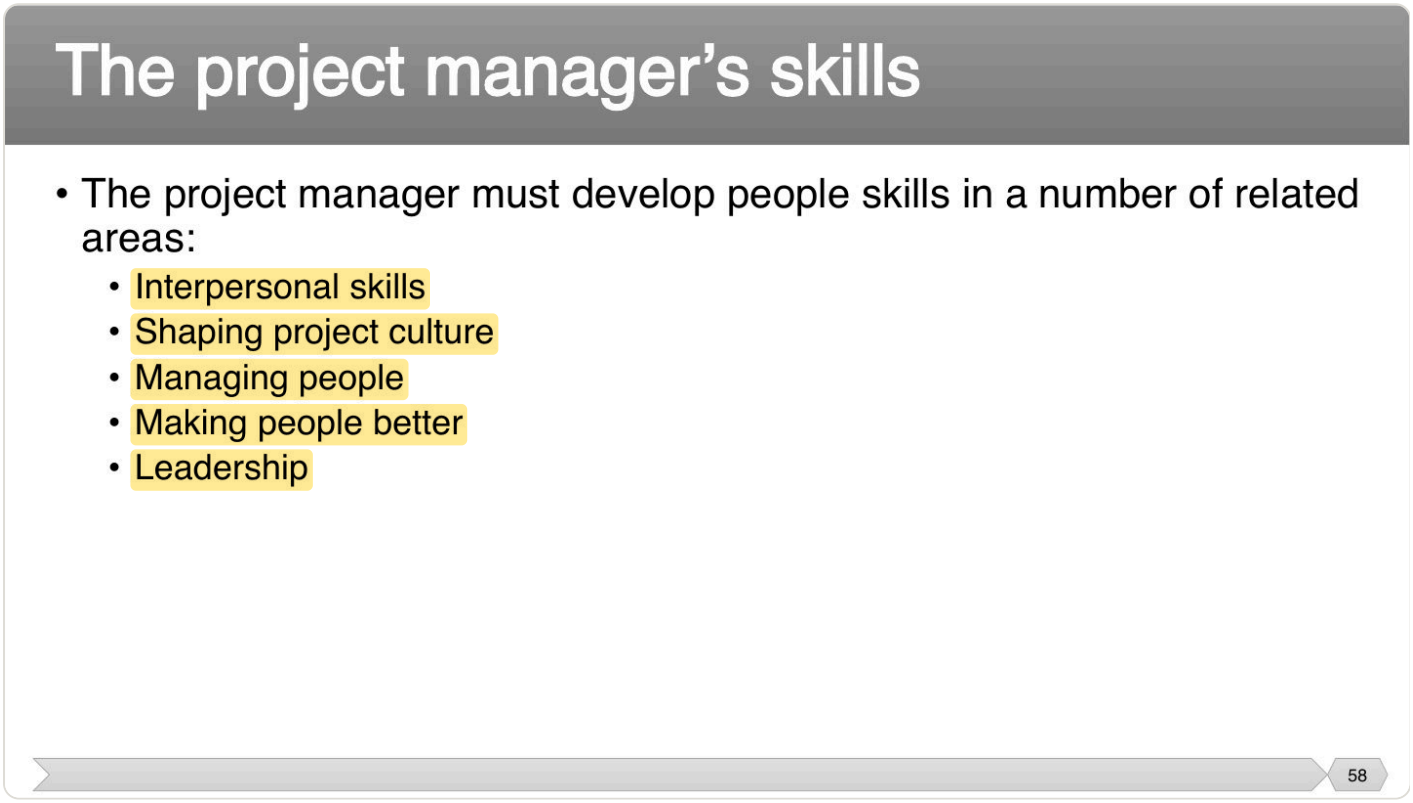
Our perspective about the team members

- The phrase ‘human resources’ has its origins as an economic and corporate asset term
- These origins lead to two semantic **problems**:
  - People are (to some extent) considered in the same light as inanimate resources: “maximizing the ROI in human capital is the goal of HR management”
  - People are considered (to some extent) as being uniform, interchangeable, and expendable
- To improve the chances of project success, the project manager should ensure that he views and acts with his team members as “collaborators”, “stakeholders”, “teammates” by recognizing and nurturing their individual talent and fostering psychological safety and team cohesion.

NOTES

1. "Human resources" comes from an **economic** term for a corporate asset.
2. That causes two problems: people get treated like **inanimate** resources ("maximize ROI in human capital") and as **uniform, interchangeable and expendable**.
3. Instead, the PM should treat team members as **collaborators**, stakeholders and teammates: recognize and nurture each person's talent and foster **psychological safety** and cohesion.

SLIDE 58 / 77 **The project manager's skills**



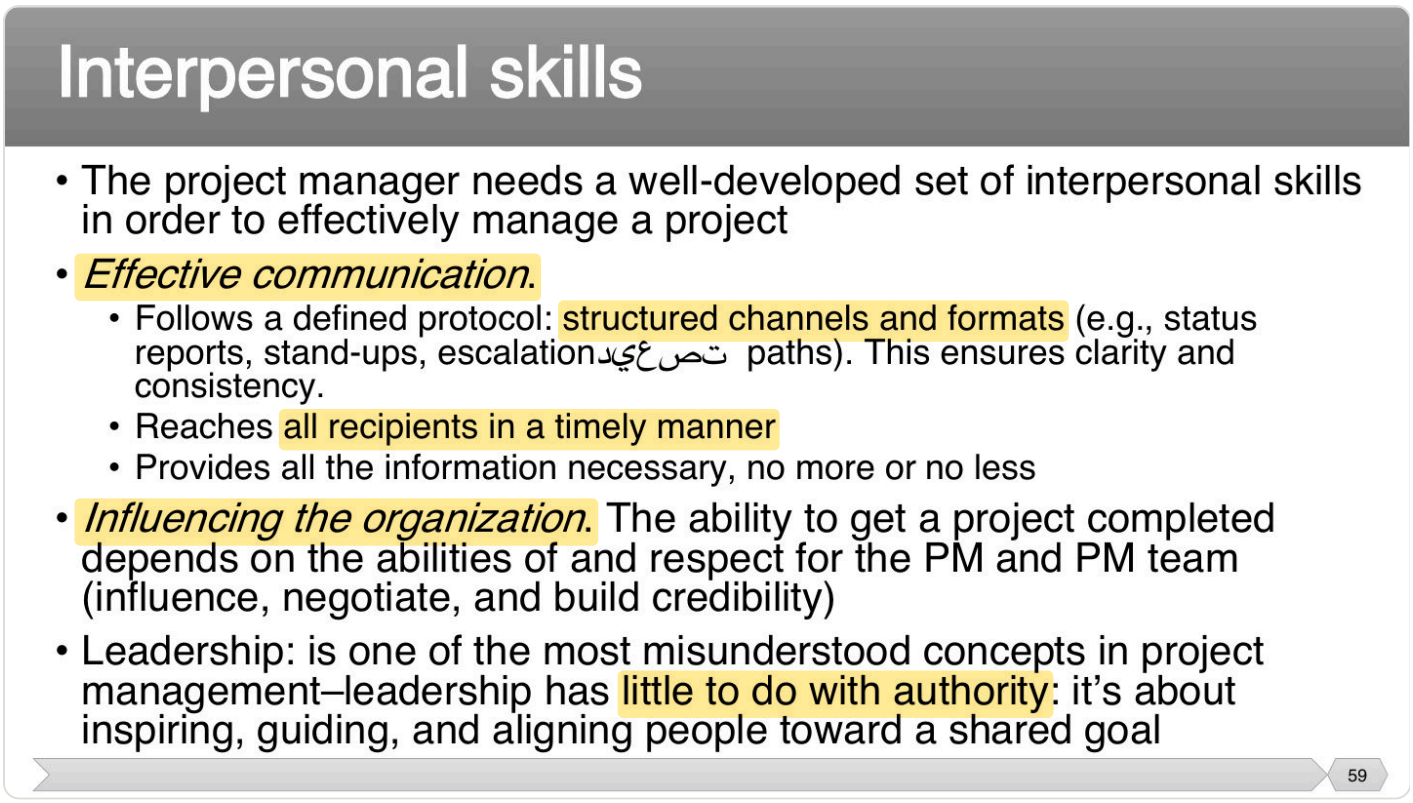
NOTES

1. The PM's people skills fall into five areas, covered in order on the next slides: **interpersonal skills** and **shaping project culture**...
2. ...**managing people** and **making people better**...
3. ...and **leadership**.

TIPS & TRICKS

- **SLiMMIng**: Shaping project culture, Leadership, Managing people, Making people better, Interpersonal skills.

 Interpersonal skills



NOTES

1. **Effective communication** follows a defined protocol (structured channels and formats: status reports, stand-ups, escalation paths), reaches **all recipients in a timely manner**, and gives all the information needed, no more and no less.
2. **Influencing the organization**: getting the project done depends on the PM's abilities and the respect they earn; they influence, negotiate and build credibility.
3. **Leadership** is one of the most misunderstood concepts: it has **little to do with authority**; it is about inspiring, guiding and aligning people toward a shared goal.

EXAM ANGLE

Past exam · Final 251 · Q3.1 — Q1

In what ways do proficient communication abilities contribute to the management of project risks and the attainment of project success? Illustrate with an instance of how a breakdown in communication could affect a project, and propose methods for prevention or resolution.

Answer: Good communication surfaces risks and issues early, while they are cheap to handle; keeps requirements and expectations shared across the team and stakeholders; and supports fast, informed decisions. Example: the backend team changes an API without telling the frontend team, integration fails at the end of the sprint, causing rework and a missed release. Prevention: a communication plan with defined channels (stand-ups, change notifications, API contracts), timely updates to everyone affected, and an escalation path; resolution: a joint meeting to agree the interface and re-plan.

Interpersonal skills (2)

Interpersonal skills (2)

- *Motivation*. Providing people with incentives to act in the interest of the project and overcome inevitable challenges
  - Motivation comes in a number of forms; studies show financial incentives are among the least effective motivators
- *Negotiation and conflict management*. Working with others to come to agreement and overcome differences, real or perceived
  - Reaching consensus and defusing conflict are two elements of these skills
- *Problem solving*. Being able to recognize and define problems; formulate, evaluate, and make decisions regarding possible solutions
  - Problem solving is a skill that can be learned but requires practice: progressive elaboration helps manage the complexity of most project problems

NOTES

1. **Motivation**: giving people incentives to act in the project's interest and overcome challenges. Studies show **financial incentives are among the least effective** motivators; recognition, growth and meaningful work do more.
2. **Negotiation and conflict management**: working with others to reach agreement and overcome differences; **reaching consensus** and defusing conflict are its two elements.
3. **Problem solving**: recognize and define problems, then formulate, evaluate and decide on solutions. It can be learned with practice; **progressive elaboration** (refining step by step) helps manage complex problems.

TIPS & TRICKS

- Slides 59 and 60 together, **IMPLEmeNt**: Influencing, Motivation, Problem solving, Leadership, Effective communication, Negotiation.

EXAM ANGLE

Prediction Q1

Studies show financial incentives are among the most effective motivators for project teams.

T True

F **False**

> The slide says they are among the least effective.

 Shaping project culture

Shaping project culture

- The project manager must shape the project culture within the broader context of the framework of the organizational culture
- *Understand organizational culture*
  - Understanding the attitude of the organization toward the project and toward project management processes is essential (Every organization has its own values, norms, and attitudes toward projects—some embrace structured methodologies, others prefer informal or ad hoc approaches)
  - Organization attitudes toward *real*–as opposed to fanciful–project management practices varies widely (practices that exist **in theory but not in reality**)
- *Understand each team member's engineering and personal background*
  - Education, experience, generation, and personality all determine what roles the person may take and how they will fit in with the rest of the team
- Assess each team member's cultural role or roles and match this to their project role

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NOTES

1. The PM shapes the **project culture** within the **broader context** of the organization's culture, so first understand the organizational culture: its values and attitude toward projects (structured methods or ad hoc).
2. Organizations differ widely in their attitude to **real** PM as opposed to "fanciful" PM: practices that exist in theory but not in reality.
3. Understand each member's **engineering and personal background**: education, experience, generation and personality decide which roles suit them.
4. Assess each member's **cultural role** (next slide) and match it to their project role.

 Shaping project culture

Shaping project culture

- Match cultural and project roles to people
- Representative cultural roles include:
  - Leader (Takes initiative, inspires others)
  - Listener/talker (Facilitates communication, bridges gaps)
  - Complainer/naysayer (who habitually expresses negative or skeptical views: Raises concerns, challenges assumptions)
  - Expert (Offers deep knowledge and credibility)
  - Charger/plodder (Pushes forward with energy)
- Significant project roles include:
  - *Project manager*. Guides the team planning and tracks progress against the plan
  - *Team leader*. Builds and maintains an effective team
  - *Development leader*. Produces a superior product
- Remain non-judgmental: every cultural role has potential value for the team

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NOTES

1. **Cultural roles**: **leader** (takes initiative, inspires), **listener/talker** (bridges communication gaps), **expert** (deep knowledge, credibility), **charger/plodder** (pushes forward with energy)...
2. ...and the **complainer/naysayer**, who habitually voices negative or skeptical views but raises concerns and challenges assumptions, often the first to spot a risk.
3. **Project roles**: the **project manager** guides planning and tracks progress; the **team leader** builds and maintains an effective team; the **development leader** produces a superior product.
4. Remain **non-judgmental**: every cultural role has potential value for the team.

TIPS & TRICKS

- Cultural roles, in slide order: **L**azy **L**ions **C**atch **E**very **C**ow. Leader, Listener or talker, Complainer, Expert, Charger or plodder.

 Project environments (context: project culture)

Project environments (context: project culture)

- Cultural and social factors
  - Projects are planned and implemented in a social and cultural context
  - Some factors that will influence a project might be: economic, educational, ethical, ethnic, and religious
  - These factors can affect the authority of the project manager
- International and political
  - As globalization expands, familiarity with local laws and customs is essential for effective project management
  - Shifting international, national, and local political factors can have dramatic influences on a project
  - Some mundane factors that will influence a project might be: time zone differences, holidays, travel requirements, teleconferencing capabilities

NOTES

1. Projects run in a **social and cultural context**: economic, educational, ethical, ethnic and religious factors can affect the **authority of the PM** (e.g. how far people accept direction from a younger manager).
2. With globalization, familiarity with **local laws and customs** is essential, and shifting political factors can dramatically affect a project.
3. Mundane factors matter too: **time zone differences**, holidays, travel and teleconferencing capabilities.

 Shaping project culture

Shaping project culture

- Monitor and manage team culture as you would manage technical issues
  - Make each person's project role clear
  - Understand each person's personality and make known your understanding of each person's social role in a positive way
  - State and maintain your view of the team
    - Example: "we are all equals in this team, but if a critical situation arises, I will make the decision."
- Recognize potential cultural role problems before they have a negative impact on the team (A dominant voice silencing others, A skeptic undermining morale)
- Solve project role problems before they have a negative impact on the project

NOTES

1. **Monitor and manage team culture** as you would manage technical issues: it is a project variable, not background.
2. Make each person's **project role clear** and show, positively, that you understand their personality and social role.
3. **State and maintain your view of the team**, e.g. "we are all equals in this team, but if a critical situation arises, I will make the decision."
4. **Recognize cultural role problems** (a dominant voice silencing others, a skeptic undermining morale) and **solve project role problems** before they harm the team or project.

Project Culture

- Monitor team members' satisfaction with their role.
  - You will need to monitor team role satisfaction throughout the project.
  - It is especially important during project launch when roles can be changed easily.
- Monitor the engineering task-to-people fit to make sure each person can perform the engineering task.
  - Role mismatches lead to project problems and schedule slip.
  - If you don't review task-to-people fit, you will be unaware of each person's ability, or lack thereof, to perform tasks.
- Monitor the cultural climate of the team as a whole.
  - Monitor project culture by watching and listening to each team member.
  - Guide the cultural views of the project by your own example.
  - You need to be able to monitor the team culture throughout the project in order to build confident people and a confident team.

NOTES

1. Monitor members' **satisfaction with their role** throughout the project, especially **during launch** when roles are easy to change.
2. Monitor the **task-to-people fit**: can each person actually do their engineering tasks? **Role mismatches** cause project problems and schedule slip, and without reviewing fit you won't know.
3. Monitor the **cultural climate** of the whole team by watching and listening, and guide it by your own example, to build confident people and a confident team.

 Managing People

Managing People

- The main problems of our work are not so much technological as sociological in nature
- (optional reading) One of the most enlightened sources for information about managing people is:
  Peopleware: Productive Projects and Teams, Third Edition,
- Most technical managers come from technical backgrounds and manage that way as if they are solving an engineering problem
- Most managers do not consider individual idiosyncrasies (personal strange habit), yet these can make or break a project

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NOTES

1. The main problems in our work are **sociological**, not technological.
2. **Peopleware** (DeMarco and Lister) is the recommended optional reading on managing people.
3. Most technical managers manage as if solving an **engineering problem** and ignore **individual idiosyncrasies**, yet these can make or break a project.

 Managing People

Managing People

Guidelines for Managing People:

- Gain visibility without micromanagement.
- Review process and products, not people.
  - Say: “This testing regime allows too many bugs to slip through to the customer; what can we do to correct this?”
  - Not: “You all don’t seem to know how to write unit tests properly.”
- Coordinate, don’t manipulate.
- Use your knowledge, not your position of power.
- Channel people, don’t put dams in front of them.
  - Redirect misplaced energy or effort, don’t cut it off
- Focus on project and people needs, not your authority as manager.
  - This will maintain your leadership role without explicit effort

NOTES

1. **Gain visibility without micromanagement**: know the status through boards and reviews, not by watching over shoulders.
2. **Review process and products, not people**. Say “this testing regime lets too many bugs through to the customer; what can we do?”, not “you don’t know how to write unit tests.”
3. **Coordinate, don’t manipulate** (use knowledge, not position), and **channel people** rather than damming them: redirect misplaced energy instead of cutting it off.
4. **Focus on project and people needs**, not your authority; that maintains your leadership without explicit effort.

EXAM ANGLE

Prediction Q1

Which statement follows the guideline “review process and products, not people”?

A “You don’t know how to write unit tests.”

B “Whoever wrote this is careless.”

C “I’ll decide, since I’m the manager.”

D **“This testing regime lets too many bugs through; how can we fix it?”**

 production line efficiency

How NOT to manage 'thought workers': production line efficiency

- The application of **production line** efficiency principles to thought workers can be a sign of disaster for a project
- Common errors include:
  - Squeeze out error – make the 'machine' run as smoothly as possible
  - Take a hard line about people goofing off on the job (Assuming that any non-visible output equals laziness: creative people need time to reflect)
  - Treat workers as interchangeable pieces of the machine
  - Optimize the steady state – don't worry about start up and shut down
  - Standardize procedure – run by the book (rigid processes)
  - Eliminate experimentation (Discouraging trial-and-error or creative risk-taking)
- Consider how each of these errors might impact your own work

NOTES

1. Applying **production-line** efficiency to thought workers can be a sign of disaster. The common errors:
2. **Squeeze out error** (expecting the "machine" to run perfectly) and taking a hard line on "**goofing off**", assuming any non-visible output is laziness when creative people need time to reflect.
3. Treating workers as **interchangeable pieces** and **optimizing the steady state** while ignoring start-up and shut-down (ramp-up of new people, handover at the end).
4. **Standardizing procedure** (run everything by the book) and **eliminating experimentation**, which kills the creativity software needs.

 Making People Better

Making People Better

- Make professional development a project goal
- Recognize long- and short-term development goals
  - Short-term goals focus on skills needed for the project
  - Long-term goals prepare team members for future projects
- Let each team member specify personal improvement goals
  - Starting point: Competency framework goals such as: Technical, leadership, domain, general/personal, communication, and management
- For a fixed interval, have team members track their individual time expenditure in detail
  - This concept goes well beyond the weekly timesheet concept
  - This is an awareness-building activity, *not* a monitoring activity
  - The *Personal Software Process (PSP)* is a formal application of this idea

NOTES

1. Make **professional development a project goal**, not an afterthought.
2. **Short-term** goals focus on skills needed for this project; **long-term** goals prepare members for future projects.
3. Let each member set **personal improvement goals**, starting from a competency framework (technical, leadership, domain, personal, communication, management).
4. For a fixed period have members track their time in detail. It is an **awareness-building** activity, **not monitoring**. The **Personal Software Process (PSP)** formalizes the idea.

SLIDE 70 / 77 **Making People Better**

Making People Better

g

- *Have team members track their individual time*
  - Goes beyond the weekly timesheet concept
- One formal approach to this is the *Personal Software Process* (PSP) which shows developers how to:
  - Manage the quality of their projects
  - Make commitments they can meet
  - Improve estimating and planning
  - Reduce defects in their products

[See note below for more info.]

- One element of PSP is to track daily time spent on *all* project-related tasks:
  - Reading
  - Writing
  - Meeting
  - Training
  - Requirements
  - Design
  - Coding
  - Testing
  - Deployment
  - E-mail
  - Phone

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NOTES

1. PSP has developers track **daily time** on every project activity (reading, writing, meetings, training, requirements, design, coding, testing, deployment, e-mail, phone), beyond the weekly timesheet.

2. It shows developers how to **manage quality**, **make commitments they can meet**, **improve estimating and planning**, and **reduce defects**, because they learn from their own real data.

SLIDE 71 / 77 **Leadership**

Leadership

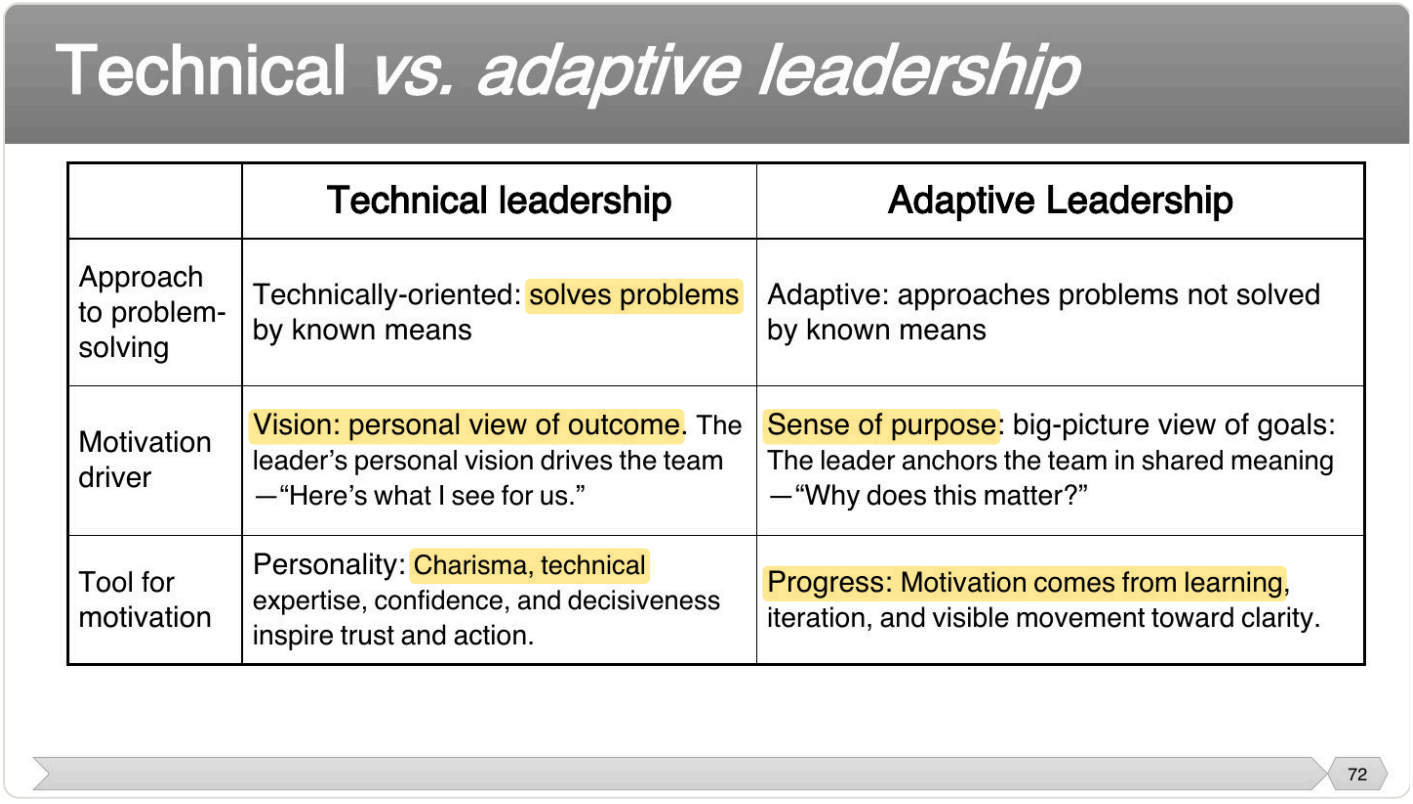
- *Leadership* is the capacity to mobilize a group to solve the most challenging problems (wrong to think that leadership is about authority)
- Two types of leadership:
  - *Technical leadership* (a.k.a. *authority*) motivates people when the problem can be solved by *known* means: the leader directs, instructs and applies best practices.
  - *Adaptive leadership* motivates people to solve problems that *cannot* be solved by known means (needed when the problem is **complex or novel**)
    - Requires learning on the part of the leader as well as the group (facilitate learning and research)
    - Often involves leading with good questions (provoke thinking, guide inquiry)
    - Requires a deep clarity of purpose (to keep the team anchored toward its goals) : Example: during brainstorming meetings, the leader is the one that needs to stop the conversation from diverging to unimportant arguments, problems and bring the discussion back to the essential questions.

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NOTES

1. **Leadership** is the capacity to **mobilize a group** to solve its most challenging problems; it is wrong to think it is about authority.
2. **Technical leadership** (a.k.a. authority) works when the problem can be solved by **known means**: the leader directs, instructs and applies best practices.
3. **Adaptive leadership** is needed when the problem can't be solved by known means (complex or novel). It requires **learning by the leader** as well as the group, and often means **leading with good questions** that provoke thinking.
4. It also needs **deep clarity of purpose** to keep the team anchored, e.g. in brainstorming the leader stops the discussion drifting into unimportant arguments and brings it back to the essential questions.

SLIDE 72 / 77 Technical vs. adaptive leadership



NOTES

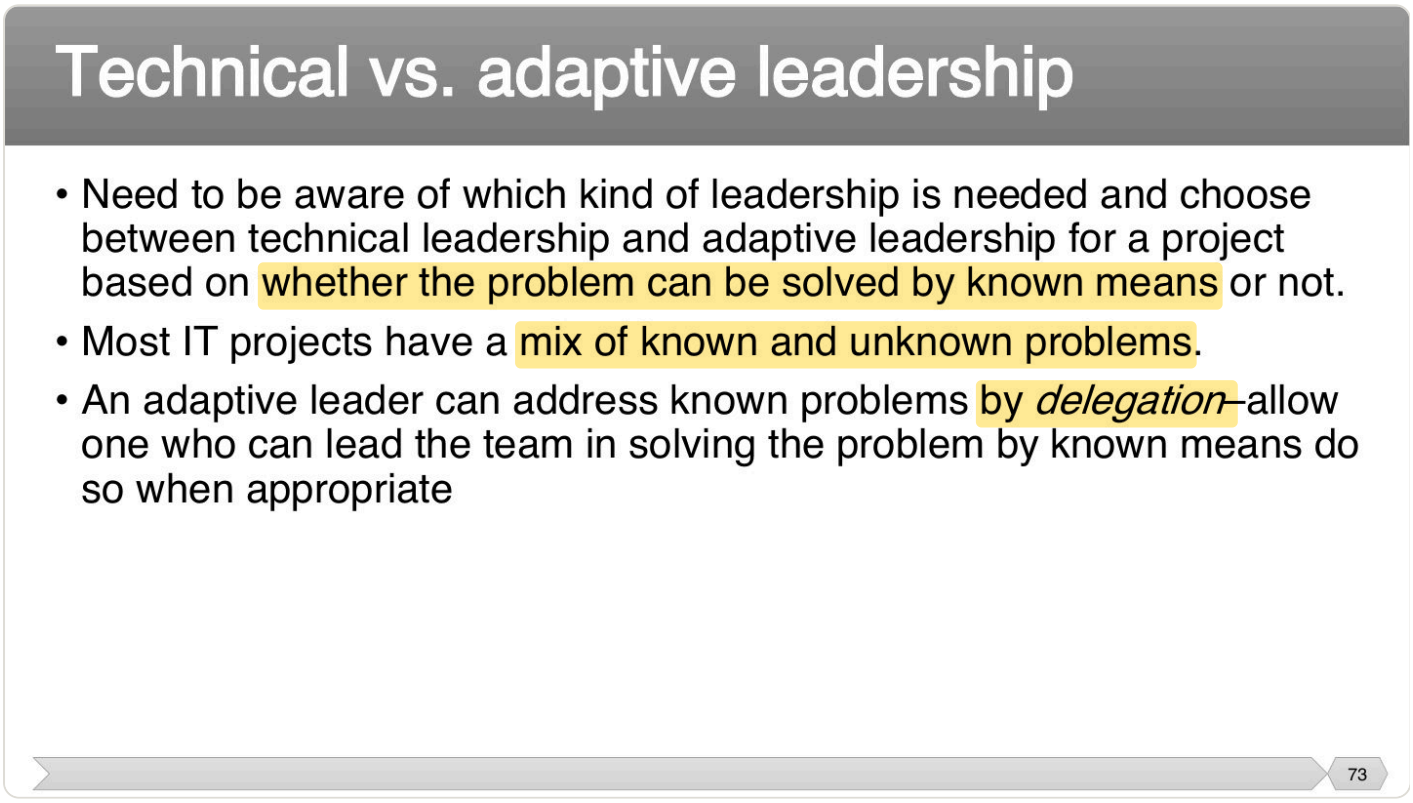
1. **Approach**: technical leadership solves problems by known means; adaptive leadership approaches problems that known means can't solve.
2. **Motivation driver**: technical uses the leader's **personal vision** ("here's what I see for us"); adaptive uses a shared **sense of purpose** ("why does this matter?").
3. **Tool for motivation**: technical relies on **personality**: charisma, expertise, confidence, decisiveness. Adaptive relies on **progress**: learning, iteration and visible movement toward clarity. In short: technical = "follow me, I know"; adaptive = "let's figure it out together."

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q5 (also V2 Short Q5) Q1

How can a project manager balance technical leadership with adaptive leadership in a software team?
Answer: First diagnose the problem: can it be solved by known means? When it can (a coding standard, a deployment the team has done before), use technical leadership: give direction from expertise and vision, or delegate it to someone who can lead it. When it can't (new technology, unclear requirements), use adaptive leadership: frame the purpose, lead with questions, let the team experiment and learn, and motivate through visible progress. Most IT projects mix both, so the PM switches between the two as the situation changes, delegating the known parts and focusing on the unknown ones.

SLIDE 73 / 77 Technical vs. adaptive leadership



NOTES

1. Choose the leadership type by asking **whether the problem can be solved by known means**.
2. Most IT projects have a **mix of known and unknown** problems, so a PM needs both.
3. An adaptive leader can handle known problems **by delegation**: let someone who can lead the team through the known solution do so.

EXAM ANGLE

Prediction Q1

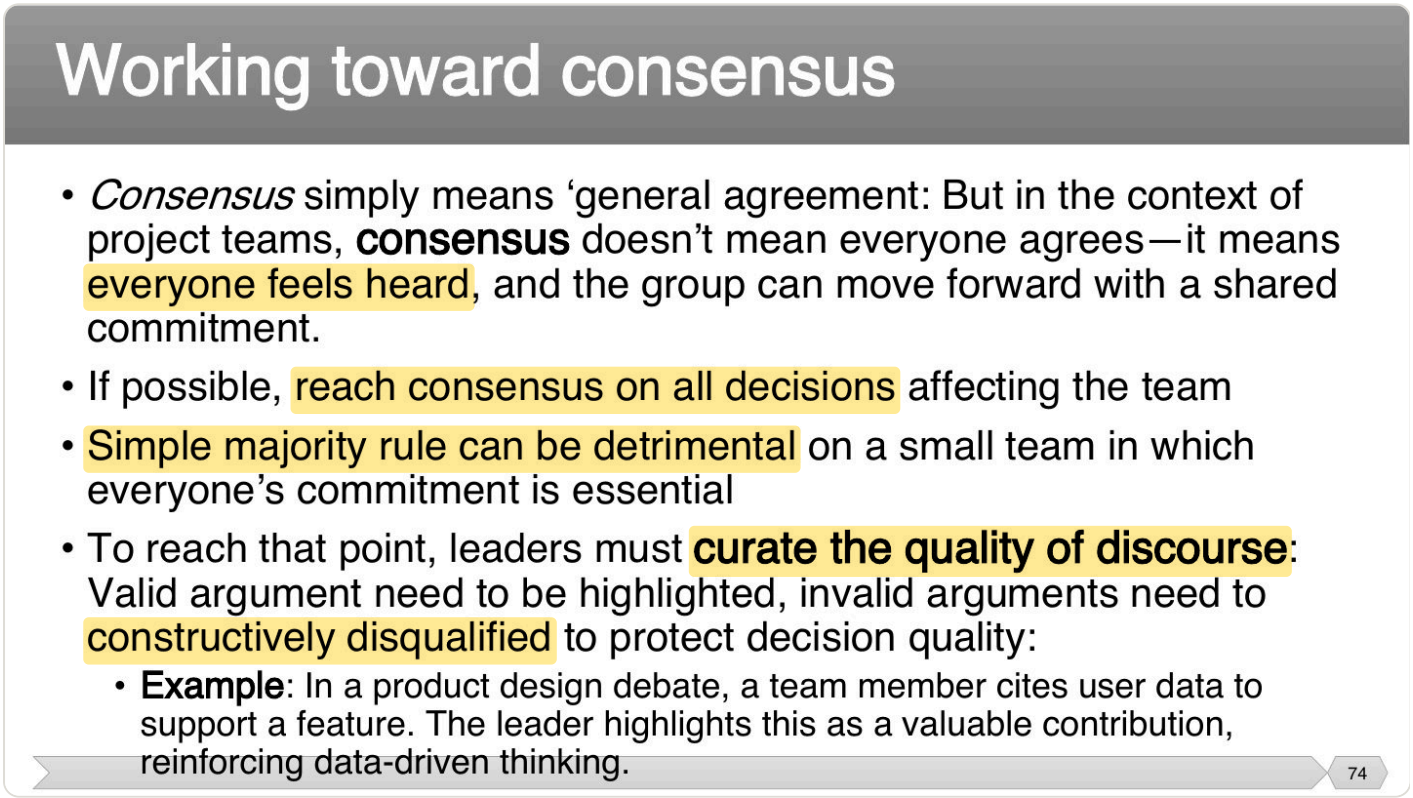
An adaptive leader should personally solve all known technical problems.

T True

F **False**

> They can address known problems by delegation.

SLIDE 74 / 77 **Working toward consensus**



NOTES

1. **Consensus** in a project team doesn't mean everyone agrees; it means **everyone feels heard** and the group moves forward with shared commitment.
2. Reach consensus on all decisions affecting the team where possible. **Simple majority rule can be harmful** on a small team where everyone's commitment is essential: the outvoted minority disengages.
3. The leader must **curate the quality of discourse**: highlight valid arguments (e.g. someone citing user data) and **constructively disqualify** invalid ones to protect decision quality.

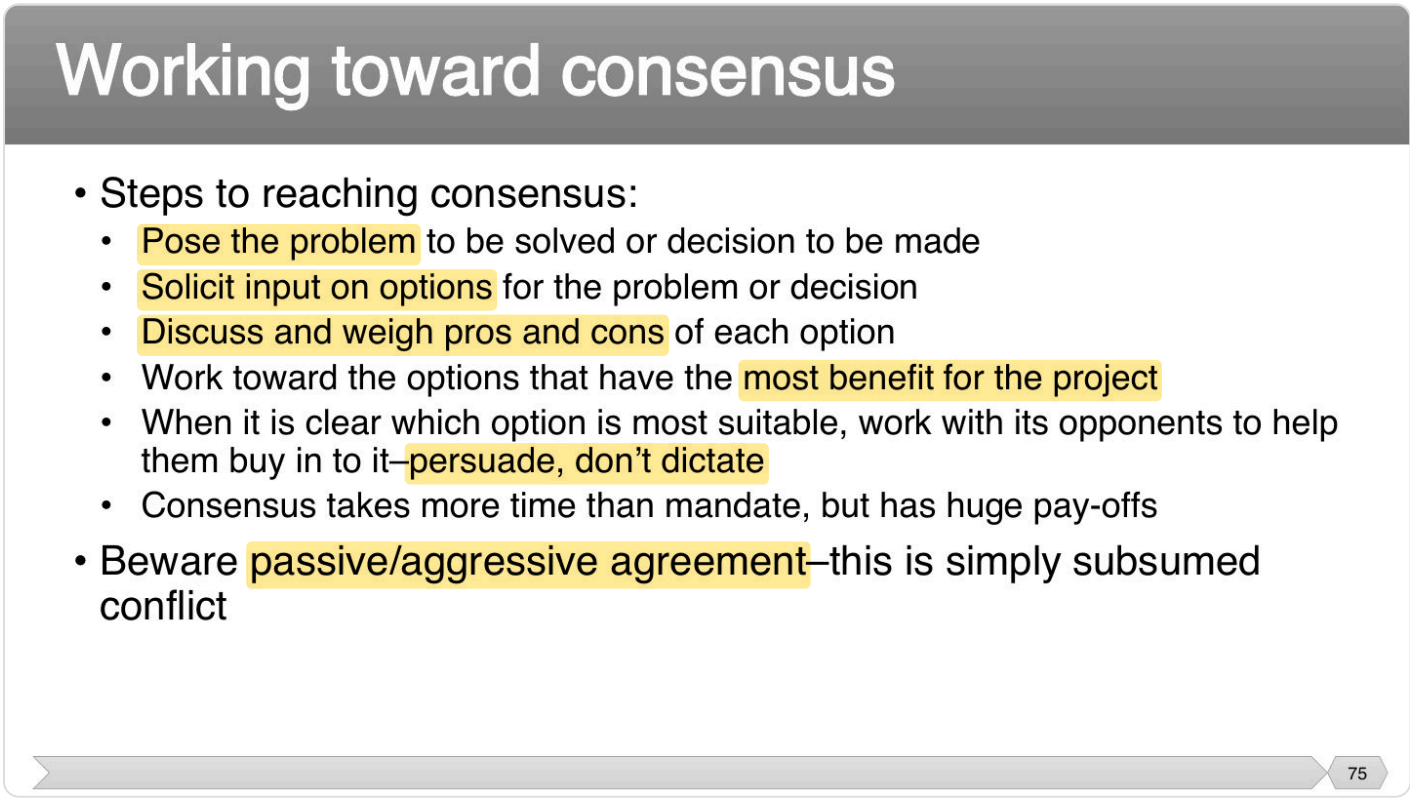
EXAM ANGLE

Past exam · Final 251 · Q5.1 Q1

Share an experience from a software project you were involved in where collaboration among team members significantly contributed to its success. Describe the specific strategies or techniques you employed to foster collaboration.

Answer: Any honest experience earns marks if it names concrete techniques. For example: in a course project building a booking app, we split the work by feature with clear owners, met twice a week for short stand-ups, tracked tasks on a shared board, used pull requests so every change was reviewed by a teammate, and paired up on the hardest integration work. Decisions were discussed until everyone had been heard, so we committed to them even when we disagreed.

SLIDE 75 / 77 **Working toward consensus**



NOTES

1. The steps: **pose** the problem or decision → **solicit input** on options → **discuss and weigh pros and cons** of each → work toward the options with the **most benefit for the project**.
2. When the best option is clear, work with its opponents to help them buy in: **persuade, don't dictate**. Consensus takes longer than a mandate but pays off in commitment.
3. Beware **passive-aggressive agreement**: people who say yes but don't act on it. It is just hidden (subsumed) conflict.

TIPS & TRICKS

- The order matters here: **P**lease **S**it **D**own, **W**e **C**onfirm. Pose the problem, Solicit input, Discuss pros and cons, Work toward the best options, Confirm the choice.

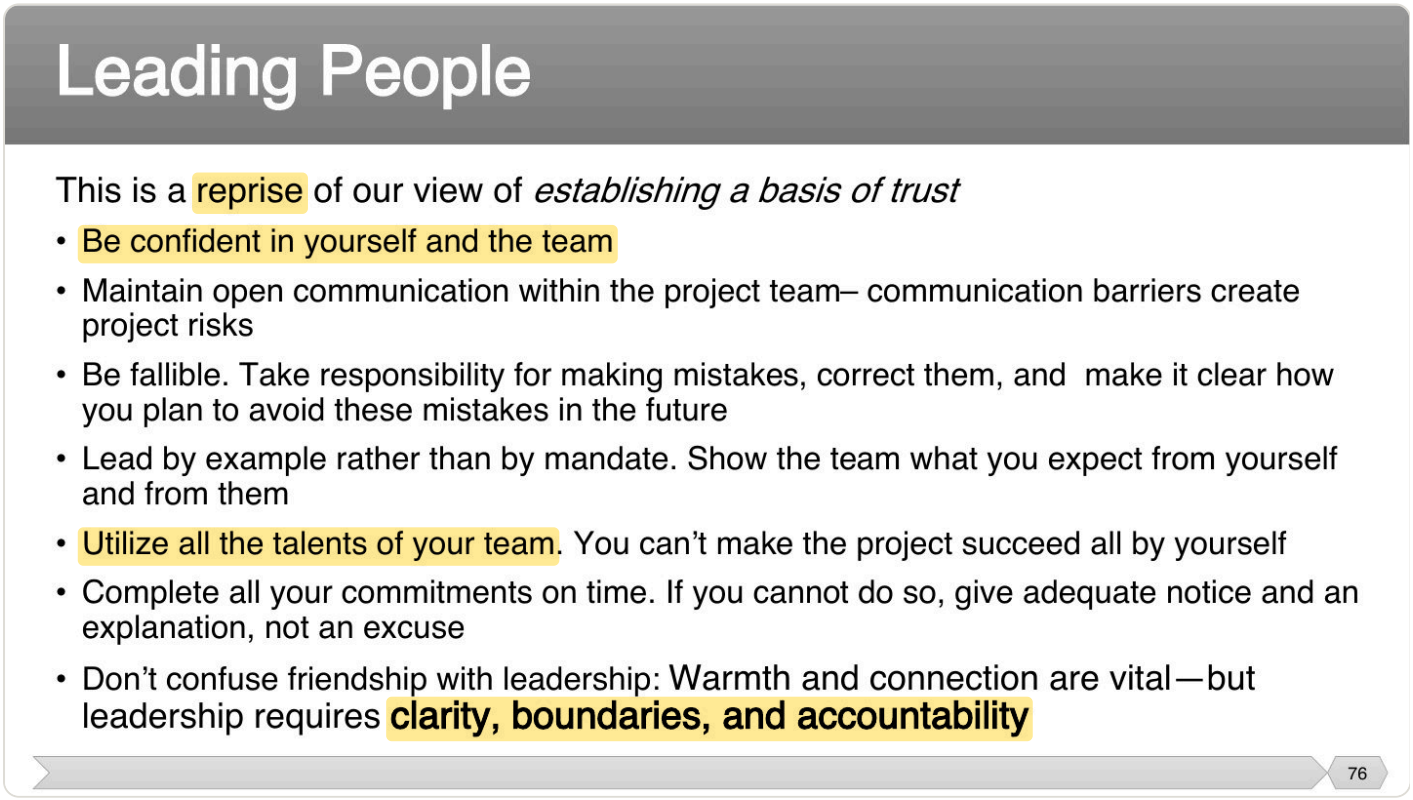
EXAM ANGLE

Past exam · Final 251 · Q5.2 Q1

You are leading a cross-functional team tasked with selecting a continuous integration/continuous deployment (CI/CD) platform for an upcoming project. The team includes software developers, DevOps engineers, and QA testers, each with distinct expectations regarding automation workflows and pipeline management. Describe your approach to collaboratively choosing the most suitable CI/CD platform.

Answer: Run the consensus steps. Pose the decision in a kickoff workshop where developers, DevOps engineers and QA testers each list their needs (build speed, deployment control, test automation). Agree the criteria and their weights together: integration with the repository and tools, automation and test support, pipeline flexibility, scalability, security, cost, ease of use and vendor support. Shortlist options, weigh each against the criteria (a weighted scoring table, or a short proof of concept), converge on the one with the most benefit for the project, then work with those who preferred others so they buy in. Don't just vote or dictate, and watch for passive-aggressive agreement.

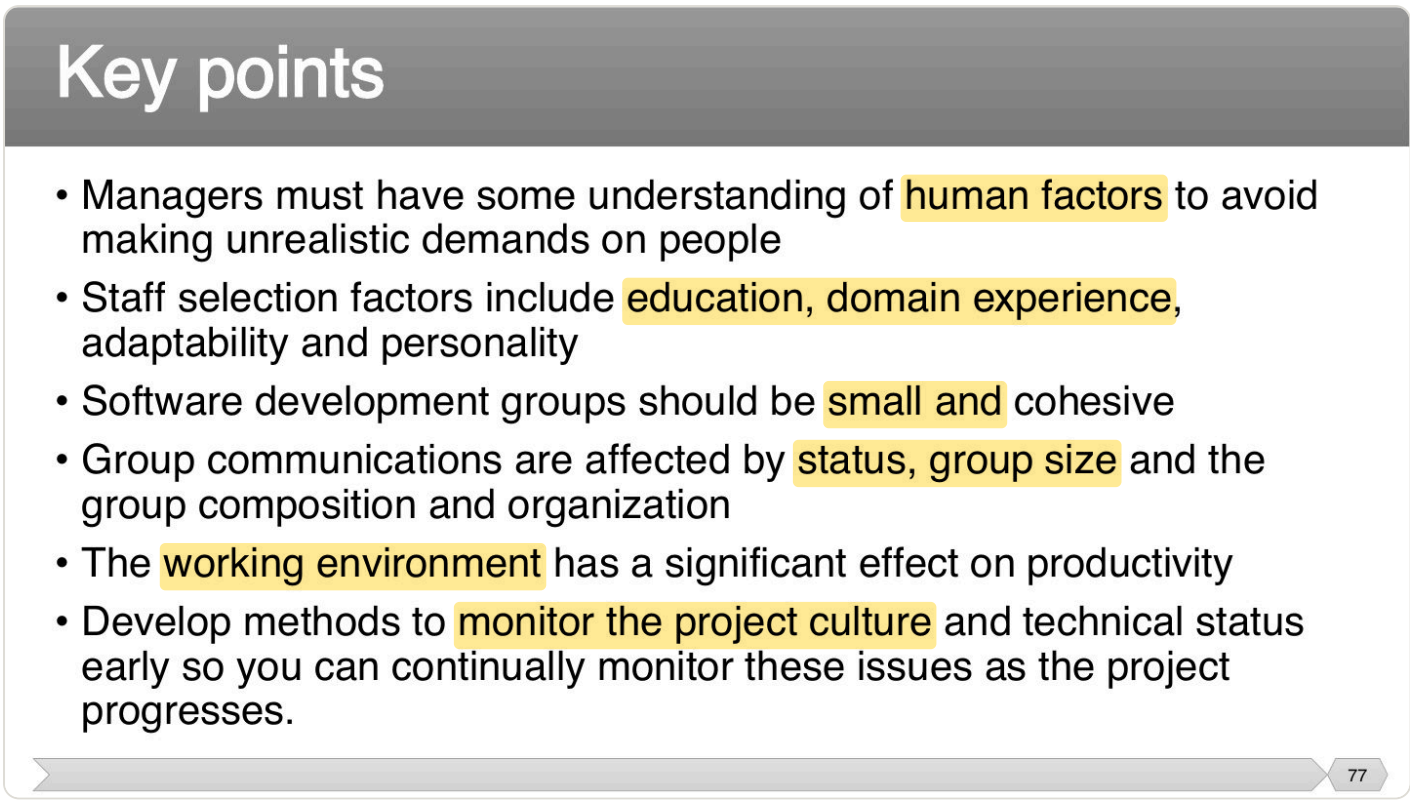
SLIDE 76 / 77 **Leading People**



NOTES

1. A **reprise** of slide 12 on establishing trust, now framed as how to lead.
2. Two additions: **be confident** in yourself and the team, and **use all the talents** of your team, since you can't make the project succeed alone.
3. Friendship ≠ leadership: warmth and connection are vital, but leadership requires **clarity, boundaries and accountability**.

 Key points



NOTES

1. Managers need to understand **human factors** to avoid unrealistic demands.
2. Staff selection factors include **education, domain experience**, adaptability and personality.
3. Groups should be **small and cohesive**; group communication is affected by **status, group size**, composition and organization.
4. The **working environment** affects productivity, and the PM should **monitor the project culture** and technical status early and throughout.

Original slides from the SE423 lecture deck, with study notes. Check exact wording against the original deck.
Notes and questions are study aids. Made by Shoug Alomran.